When Statutory References Break Semantic Retrieval: Diagnosing Extractive Question Answering over Indian Intellectual Property Statutes

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Abstract

Statutory question answering (QA) requires retrieving the governing passage and extracting the exact legal clause. We diagnose the gold-context-to-end-to-end performance gap on IndicIPR-QA, a 4,462-pair benchmark over seven Indian intellectual property Acts. The reader is not the dominant error source: given the governing paragraph, the strongest supervised encoder reaches 82.33 F1, but end-to-end F1 falls to 30.00. BM25 ranks the governing paragraph in the top five for only 22.67% of questions, while zero-shot dense retrieval and cross-encoder re-ranking reach at most 30.23%, bounded by first-stage recall. Half the test set, 177 of 344 questions, identifies its target by citation rather than content, scoring 88.63 F1 with the gold paragraph but 16.80 end to end. A non-learned regular-expression mapping from cited sections to passage metadata raises recall@5 from 22.67% to 86.34%. Replacing the cited section with another section of the same Act, leaving every content word unchanged, collapses recall@5 to 0.00% on citation-specified questions, while citation-only queries retain 62.71%: the reference carries most, but not all, of the retrievable signal. This is a diagnosis, not a retrieval advance, and depends on 94.48% of test questions carrying a parsable citation, a benchmark property rather than a claim about statutory QA generally. A size-matched three-seed ablation finds mean F1 spanning 0.08 across authentic, synthetic, and mixed text against a pooled within-corpus SD of 0.53, with no detectable corpus effect ($\boldsymbol{p = 0.98}$). Resolving statutory references and separating citation-specified from content-bearing queries should take priority over the retrieval and pretraining approaches tested here.

Keywords: Legal NLP, Extractive question answering, Legal information retrieval, Indian IPR law, Benchmark design, Domain-adaptive pretraining

1 Introduction

Indian intellectual property law is spread across seven separate statutes rather than a single unified code. Patents (1970), trade marks (1999), copyright (1957), designs (2000), geographical indications (1999), semiconductor integrated circuits layout-designs (2000), and plant varieties and farmers' rights (2001) are each governed by a separate Act. These Acts differ in their vocabulary, procedures, exceptions, and internal cross-references. As a result, finding the exact provision for questions on patent eligibility, trademark registration, or plant-variety rights is time-consuming and error-prone, even for lawyers, law students, and policy researchers.

Automated question answering (QA) can reduce this workload, but legal QA is more challenging than general-domain QA. Statutes are long and complex, and the system must extract the exact answer span rather than return a related paragraph. In IPR law, omitting a single condition or exception can change the legal meaning of an answer, while important details often appear in provisos and sub-clauses attached to the main clause. Answers in IndicIPR-QA average 35.6 words, about ten times longer than those in SQuAD (Rajpurkar et al. 2016), making precise span boundary prediction substantially more difficult. Existing QA benchmarks do not address this setting. General-domain datasets such as SQuAD use short answers from encyclopedic text, CUAD (Hendrycks et al. 2021) focuses on contract clauses, and JEC-QA (Zhong et al. 2020) uses a multiple-choice format. None supports extractive span prediction over Indian IPR statutory text.

The experiments use IndicIPR-QA, a 4,462-pair extractive QA benchmark covering all seven Indian IPR Acts, with section-level split control and verbatim answer spans. The dataset is introduced in a companion manuscript currently under review and is not a contribution of this paper. Its key statistics and validation are summarised in Section 3.1 to keep this paper self-contained, and the reproduced tables are clearly identified. Using this benchmark, we study a single question: why does extractive QA over Indian IPR statutes perform well when the governing paragraph is supplied, yet collapse when the system must retrieve that paragraph itself? A compact encoder reaches 82.33 F1 given the gold context but 30.00 F1 end to end, and this paper is an investigation of that gap. Compact encoders are the object of study because large prompted LLMs are expensive, may paraphrase instead of copying the exact statutory span, and can generate legally plausible text that does not match the source statute, whereas compact encoder models are computationally efficient, deterministic, and restricted to predicting verbatim spans, making them well suited to legally verifiable QA. However, their performance may depend on how well their pretraining and supervision match the specialised vocabulary and complex clause structures of Indian IPR law.

One plausible explanation for that gap is that the encoder is insufficiently adapted to statutory language, so this paper tests domain adaptation first. A seed-guided synthetic domain-adaptive pretraining (DAPT) pipeline for InLegalBERT extracts 5,859 candidate legal seeds from the training corpus, retains the top 1,500, and uses Gemma 3 (Gemma Team 2025) to generate 13,211 accepted statutory-style passages. Authentic-only, synthetic-only, and mixed DAPT corpora are then evaluated through SQuAD v1 transfer followed by final IPR fine-tuning. The archived composition of each corpus is reported in Table 5. Under size-matched, multi-seed control this explanation does not survive: corpus composition has no effect that the experiment can detect. The pipeline is therefore reported as an eliminated candidate rather than as a proposed method, and the investigation moves to retrieval, where the evidence turns out to lie.

This paper makes three contributions:

1. A diagnosis of the dominant retrieval failure on IndicIPR-QA, showing that standard lexical and semantic retrieval fails particularly on citation-specified questions because it does not resolve statutory references. BM25, four zero-shot dense retrievers, sparse–dense fusion, and a cross-encoder re-ranker are compared end to end; the best of them reaches 30.23% recall@5 against a first-stage ceiling of about 42%. A citation-aware retriever that indexes section numbers, and contains no learned component, instead reaches 86.34% recall@5 and 95.64% recall@10. Ablations removing, isolating, and corrupting the statutory reference establish that the citation-aware mechanism depends causally on that reference: corrupting the cited section number collapses recall@5 from 83.62% to 0.00% on citation-specified questions, whereas first-stage content-based retrieval shows no statistically detectable change. Explicit statutory references therefore form a distinct and dominant structural retrieval signal, while content matching still does necessary work in that pipeline: a cited number occurs in more than one Act for 92.6% of matched queries, so once the citation narrows the candidates, content is what selects the governing statute.
2. A benchmark-design finding with consequences for how statutory QA is measured. Half of IndicIPR-QA, 177 of 344 test questions, identifies its target by citation rather than by content. These items are substantially easier for gold-context span extraction but substantially harder for ordinary retrieval, which explains the per-Act inversion in which the highest-scoring Act under gold context is the hardest to retrieve. The headline gold-context score of 82.33 F1 decomposes into 88.63 on the citation-specified half and 75.65 on the content-bearing half.
3. A controlled non-detection for seed-guided synthetic domain-adaptive pretraining. With corpus scale, record multiplicity, and checkpoint selection controlled, we detect no reproducible advantage of synthetic over authentic text under the present three-seed design; because the confidence intervals remain wide, this is a non-detection rather than evidence of equivalence. Synthetic text does not outperform authentic text, and composition, scale, multiplicity, and checkpoint rule are each worth less than the seed-to-seed standard deviation. Repeating the full pipeline under three seeds for each size-matched corpus turns the composition bound into a tested non-detection: 0.08 F1 between corpus means against a pooled within-corpus standard deviation of 0.53, with no detectable corpus effect under analysis of variance ($p = 0.98$). The pipeline and its audited corpora are reported in full so that the negative result is checkable.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the benchmark and the seed-guided synthetic DAPT pipeline used to test the domain-adaptation hypothesis. Section 4 presents the experimental setup, and Section 5 reports the experimental results, including model comparisons, ablations, and diagnostic analyses. Section 6 examines the generalisability of the findings, while Section 7 discusses their implications, limitations, and directions for future work. Section 8 concludes the paper.

2 Related Work

Work related to this paper falls into three areas. The first covers extractive QA and legal benchmarks, highlighting the lack of a benchmark for extractive QA over Indian IPR statutes. The second reviews domain-adaptive pretraining and synthetic data, which form the basis of the adaptation pipeline tested here. The third examines retrieval for legal QA, motivating the end-to-end retrieval study presented in this paper.

2.1 Extractive QA and Legal Benchmarks

General-domain extractive QA benchmarks established the standard task format, but they were developed for encyclopedic or exam-style text rather than long statutory provisions. SQuAD 1.1 (Rajpurkar et al. 2016) introduced extractive QA with short Wikipedia answer spans, while SQuAD 2.0 (Rajpurkar et al. 2018) added unanswerable questions. Natural Questions (Kwiatkowski et al. 2019), TriviaQA (Joshi et al. 2017), HotpotQA (Yang et al. 2018), and RACE (Lai et al. 2017) expanded QA to search queries, distant supervision, multi-hop reasoning, and exam-style reading comprehension. Indic-QA (Singh et al. 2024) further introduced a multilingual benchmark across 11 Indian languages. However, none of these datasets targets extractive QA over Indian statutory text.

Legal NLP has introduced several domain-specific benchmarks, but none focuses on extractive span prediction over Indian IPR statutes with section-level split control. CUAD (Hendrycks et al. 2021) targets commercial contracts, JEC-QA (Zhong et al. 2020) uses multiple-choice questions for Chinese law, and LegalBench (Guha et al. 2023), LexGLUE (Chalkidis et al. 2022), and LawBench (Fei et al. 2024) cover legal reasoning tasks across American, European, and Chinese jurisdictions. Statutory QA datasets such as Chinese EQUALS (Chen et al. 2023) and Vietnamese VLQA (Nguyen et al. 2025) further demonstrate that statutory QA requires jurisdiction-specific resources. Similarly, IPBench (Wang et al. 2026) and IPEval (Wang et al. 2024) evaluate intellectual property knowledge but do not support extractive QA over Indian IPR statutes. Closest in setting, DiscoLQA (Sovrano et al. 2024) performs zero-shot question answering over European legislation by exploiting the discourse structure of legal language, but it targets EU regulations rather than verbatim span extraction from Indian IPR statutes.

Indian legal NLP has primarily focused on court-centric tasks rather than statutory QA. ILDC (Malik et al. 2021) established a benchmark for judgment prediction, while Paul et al. (2023) showed that continued pretraining on Indian legal corpora improves downstream legal NLP. Those encoders form the basis of our experiments. More recently, Nigam et al. (2023) demonstrated that QA pipelines for Indian criminal law remain limited by retrieval quality, and InLegalLLaMA (Nigam et al. 2025) showed the benefits of instruction tuning for Indian legal reasoning. IndicLegalQA (Veningston and Mishra 2025) introduced a QA dataset from Indian court judgments, but it focuses on judicial proceedings rather than statutory span extraction. Recent work continues this court-centric focus. Prabhakar and Pati (2025) classify Indian judgments into legal domains using embeddings, feature selection, and ensemble strategies, while Deroy et al. (2024) evaluate large language models for summarising Indian and UK Supreme Court judgments and report hallucinations and inconsistencies in the generated summaries. The latter finding supports the extractive formulation adopted here, in which every answer is a verbatim span of the source statute and can therefore be checked against it.

Overall, existing benchmarks either target general-domain QA, other legal jurisdictions, or Indian court judgments. To the authors' knowledge, IndicIPR-QA is the first benchmark for extractive QA over Indian IPR statutes, combining section-level split control with verbatim answer spans averaging 35.6 words.

2.2 Domain-Adaptive Pretraining and Synthetic Data

Domain-adaptive pretraining (DAPT) has consistently improved encoder performance in specialised domains, and it motivates the first explanatory hypothesis tested in this work. Gururangan et al. (2020) showed that continued in-domain masked language model (MLM) pretraining improves downstream performance, a finding later confirmed by domain-specific encoders such as SciBERT (Beltagy et al. 2019), BioBERT (Lee et al. 2020), and LEGAL-BERT (Chalkidis et al. 2020). At the decoder scale, SaulLM (Colombo et al. 2024) demonstrated the benefits of continued legal pretraining and instruction tuning for generative legal LLMs. The work most closely related to ours is TOP-Training (Sengupta et al. 2025), which extracts domain-specific entities, generates synthetic contexts, and performs continued encoder pretraining for low-resource medical extractive QA. Similarly, Bashir et al. (2026) showed that controlled synthetic data generation with rule-based filtering improves legal domain adaptation for German NLP tasks.

Synthetic data generation has also emerged as an effective strategy for low-resource NLP. Lewis et al. (2019) demonstrated that cloze-style question generation from unlabelled text enables unsupervised QA, while West et al. (2022) used LLMs for symbolic knowledge distillation to construct commonsense datasets. Self-Instruct (Wang et al. 2023) further showed that bootstrapping instruction-following data from model-generated outputs, combined with quality filtering, improves instruction tuning without human annotation. Similarly, GPT-4 (Achiam et al. 2023) enabled high-quality structured data generation, and Honovich et al. (2023) found that models trained on LLM-generated instructions can outperform those trained on human-authored instructions. Two legal studies are especially close to this work. Yuan et al. (2023) construct a legal question bank with a large pre-trained language model to make legal knowledge more publicly accessible, and Italiani et al. (2025) generate legal QA data by distilling knowledge from large language models, showing that compact supervised models can be trained without human annotation. Both use an LLM to produce legal training data, as done here, but neither applies the generated text to continued masked language model pretraining over statutory provisions.

These studies demonstrate the value of combining DAPT with synthetic data, but none focuses on Indian IPR law or evaluates a three-stage pipeline comprising DAPT, general extractive QA transfer, and final statutory fine-tuning. We evaluate a seed-guided synthetic DAPT pipeline adapted from the entity-grounded idea of TOP-Training to Indian IPR statutes by conditioning Gemma 3 generation on legal seed entities across multiple statutory writing styles and applying rule-based validation to ensure clause-boundary correctness.

2.3 Retrieval for Legal QA

Retrieval quality is a well-known bottleneck in QA systems, particularly for legal statutes, where many provisions share similar vocabulary. Sparse methods such as BM25 (Robertson and Zaragoza 2009) remain strong first-stage baselines but struggle with vocabulary mismatch and near-duplicate legal clauses. Dense Passage Retrieval (Karpukhin et al. 2020) showed that dual-encoder retrievers can outperform BM25 in open-domain QA, while BERT cross-encoders (Nogueira and Cho 2019) improve re-ranking at higher computational cost. Retrieval-Augmented Generation (Lewis et al. 2020) further established retrieval-reader architectures as the standard approach for knowledge-intensive NLP.

Recent legal retrieval studies highlight the unique challenges of statutory retrieval. Recent work has also begun to model statutory structure more explicitly. Alshehri et al. (2026) evaluate provision-level retrieval over UK legislation with BM25 and dense candidate pools followed by a broad set of neural re-rankers. Noce et al. (2026) encode cross-article references in JuriFindIT and combine dense encoders with a legislative graph, while Akarajaradwong et al. (2025) test hierarchy-aware chunking and cross-referencing for Thai legal QA. Chae et al. (2026) likewise use graph-guided retrieval for hierarchically linked regulatory evidence, and Han et al. (2026) treat legal citation prediction as a retrieval problem over Australian case law. Collectively, these studies establish that legal structure and citations can matter beyond topical similarity. Their focus, however, is practitioner-style provision ranking, document-side hierarchy and cross-reference structure, or citation prediction. The diagnostic here is narrower: it isolates query-side statutory references and measures how removing, isolating, and corrupting that signal changes retrieval and benchmark interpretation.

LegalBench-RAG (Pipitone and Alami 2024) argues that legal RAG requires retrieving precise statutory passages rather than broad document matches. Zheng et al. (2025) showed that legal retrieval must support downstream reasoning in addition to lexical matching, while Legal RAG Bench (Butler and Butler 2026) and Afane et al. (2026) found that retrieval quality largely determines end-to-end legal QA performance. Statutory retrieval has also been studied directly. Šavelka and Ashley (2021) treat the discovery of sentences that explain statutory terms as an ad hoc retrieval task and show that relevance there depends on legal interpretation rather than surface similarity, while van Opijnen and Santos (2017) argue that relevance in legal information retrieval is multi-dimensional and cannot be reduced to topical matching. These observations are consistent with the results reported here, where neither lexical nor semantic matching reliably identifies the governing provision. Together, these findings motivate the end-to-end retrieval evaluation presented in this paper.

Table 1 positions this work against four representative studies spanning the main dimensions examined here. Each addresses one aspect of our problem, including synthetic domain-adaptive pretraining, legal retrieval, large-scale legal adaptation, or Indian legal modelling. However, none combines extractive QA over Indian statutes with an end-to-end retrieval study while also reporting uncertainty in the observed adaptation gains.

Table 1 Scope of four representative studies along the dimensions this paper examines. ✓ indicates that a dimension is within a study's stated scope, not a judgement of quality; each work addresses questions outside this grid.

Study	Synth. DAPT	Statutory span QA	Retrieval study	Indian law	Uncertainty reported
TOP-Training (Sengupta et al. 2025)	✓				
LegalBench-RAG (Pipitone and Alami 2024)			✓		
SaulLM (Colombo et al. 2024)					
InLegalLLaMA (Nigam et al. 2025)				✓	
This work	✓	✓	✓	✓	✓

3 Benchmark and the Domain-Adaptation Hypothesis

This section describes the benchmark and the domain-adaptation pipeline whose contribution this paper sets out to measure. IndicIPR-QA, a 4,462-pair benchmark covering seven Acts, serves as the evaluation benchmark. The seed-guided synthetic domain-adaptive pretraining (DAPT) pipeline augments authentic statutory text with LLM-generated passages conditioned on legal entity seeds and trains the model in three stages. It is described in sufficient detail to establish that the null comparison in Section 5.5 concerns a fully specified and reproducible adaptation procedure rather than an under-specified baseline. The following subsections describe the benchmark, seed extraction, passage generation, quality filtering and corpus construction, and the pretraining objective. Figure 1 illustrates the complete pipeline, while Algorithm 1 summarises the procedure.

3.1 Dataset Description

IndicIPR-QA is an extractive QA benchmark containing 4,462 question–answer pairs from 1,116 paragraph-level statutory contexts across seven Indian IPR Acts: Patents, Trade Marks, Copyright, Designs, Geographical Indications, Semiconductors, and Plant Varieties. Each answer is a verbatim span from its source context, making the task one of extractive QA rather than open-ended legal inference. To prevent context leakage, the dataset is split at the statute-section level, yielding 3,710 training pairs, ipr pipeline updated.jpg

Fig. 1 Overview of the pipeline evaluated in this paper. *Left:* Benchmark construction and section-level data splitting (reproduced from the companion benchmark manuscript under review). *Centre:* Seed extraction, synthetic passage generation, corpus assembly, and masked language model (MLM) pretraining, drawing on training-split contexts (Table 5); the authentic pools were built against an earlier state of the split and additionally carry

Algorithm 1 Seed-Guided Synthetic DAPT Pipeline

Input: Training split $\mathcal{D}_{tr}$, validation split $\mathcal{D}_{va}$, archived authentic statutory context record multiset $\mathcal{C}_{tr}$ (3,710 records, one per training QA pair, over 901 distinct passages; against the final split these are 898 training contexts plus three test contexts, audited in Section 6), pilot synthetic pool $\mathcal{X}_{pil}$, generator set $\mathcal{G} = \{\text{Gemma 3:12b}, \text{ Gemma 3:27b}\}$, fixed SQuAD v1 subset $\mathcal{Q}_{SQ}$ (20k train / 2k val), base encoder E, retry budget $R = 2$

Output: Domain-adapted extractive QA model C'

Stage 1 – Seed Extraction

$\mathcal{S} \leftarrow$ RuleExtract$(\mathcal{D}_{tr}) \cup$ KeyBERTExtract$(\mathcal{D}_{tr})$ ▷ *5,859 candidate legal seeds*
$\mathcal{S} \leftarrow$ SelectTopSeeds$(\mathcal{S}, 1500)$ ▷ *Min count 3; retain top 1,500 seeds*

Stage 2 – Synthetic Passage Generation

$\mathcal{X}_{syn} \leftarrow \emptyset$
for all $g \in \mathcal{G}$ **do** ▷ *Two generation passes; per-model counts in Table 4*
  for all $s \in \mathcal{S}$ **do**
    for $k \leftarrow 1$ **to** 8 **do**
      $x \leftarrow$ GenSolo(g, s, k, R) ▷ *8 solo legal-style variants; up to R retries per slot*
      if ValidateSynth(x) **then**
        $\mathcal{X}_{syn} \leftarrow \mathcal{X}_{syn} \cup \{x\}$
      end if
    end for
    for $k \leftarrow 1$ **to** 6 **do**
      $b \leftarrow$ PickSupportSeed$(s, \mathcal{S})$ ▷ *Same or related Act if available*
      $x \leftarrow$ GenBroad(g, s, b, R) ▷ *6 broad-supported variants; up to R retries per slot*
      if ValidateSynth(x) **then**
        $\mathcal{X}_{syn} \leftarrow \mathcal{X}_{syn} \cup \{x\}$
      end if
    end for
  end for
end for
$\mathcal{X}_{syn} \leftarrow$ MergeAccepted$(\mathcal{X}_{syn})$ ▷ *30,282 attempts → 13,211 accepted after exact-text merge*

Stage 3 – Corpus Assembly

$\mathcal{L} \leftarrow \mathcal{C}_{tr} \parallel \mathcal{X}_{pil} \parallel \mathcal{X}_{syn}$ ▷ *Concatenate: 3,710 authentic records, 312 pilot, 13,211 main*
$\mathcal{X} \leftarrow$ Dedup$(\mathcal{L}, \ \kappa(x) = $ Lower$(x)[1{:}200])$ ▷ *Prefix fingerprint collapses repeated contexts and shared statutory openings* ▷ $866 + 305 + 13{,}043 = 14{,}214$ *records (Table 5)*

Stage 4 – Three-Stage Training

$E_1 \leftarrow$ MLMPretrain$(E, \mathcal{X})$ ▷ *Mixed-corpus DAPT; no validation split at this stage*
$E_2 \leftarrow$ SQuADTransfer$(E_1, \mathcal{Q}_{SQ})$ ▷ *Intermediate extractive QA transfer on fixed 20k/2k SQuAD v1 subset*
$C' \leftarrow$ IPRFineTune$(E_2, \mathcal{D}_{tr}, \mathcal{D}_{va})$ ▷ *Final fine-tuning on IndicIPR-QA*
return C'

408 validation pairs, and 344 test pairs with no shared contexts across splits. Table 2 summarises the key benchmark statistics.

Table 2 Summary statistics of the IndicIPR-QA benchmark, reproduced from the companion benchmark manuscript under review.

Acts covered	7
Sections covered	681
Paragraph-level contexts	1,116
Total QA pairs	4,462
Train / Validation / Test	3,710 / 408 / 344
Avg. context length (words)	171.3
Avg. question length (words)	11.9
Avg. answer length (words)	35.6
Median answer length (words)	28
Max answer length (words)	375
Multi-clause answers (%)	19.9
Q–ctx lexical overlap (mean)	0.448

Construction of IndicIPR-QA is described in the companion manuscript under review and is not part of this study. The dataset was created in three stages: (i) GPT-4o-based QA generation from paragraph-level statutory contexts; (ii) a nine-step quality filtering pipeline covering length constraints, span validity, question clarity, and lexical coverage; and (iii) manual curation to remove residual boundary errors. A dataset-level ablation showed that the effect of training data quality is not monotonic. Removing manual curation while retaining all automated filters reduced mean F1 by 3.80 points, although the loss was recovered with larger candidate pools. The full three-seed analysis, separating training utility from evaluation-set quality, is reported in Section 5.17.

For human validation, two annotators independently evaluated a stratified sample of 200 QA pairs for question clarity, answer correctness, and answer minimality. As summarised in Table 3, strict acceptability rates reached 99.0% for question clarity, 95.0% for answer correctness, and 75.5% for answer minimality. Agreement on answer minimality was substantial, with a quadratic-weighted Cohen's $\kappa_{qw} = 0.613$.

Table 3 Human validation of a stratified 200-pair sample, reproduced from the companion benchmark manuscript under review

Criterion	Acceptability	Agreement
Question clarity	99.0%	high (skewed)
Answer correctness	95.0%	fair κ_{qw}
Answer minimality	75.5%†	substantial κ_{qw}

†Strict rate: both annotators independently rated $\geq$2. Pooled human accept rate is 82.0%. Agreement is weighted Cohen's κ (Cohen 1968); strength labels follow (Landis and Koch 1977).

Two characteristics make IndicIPR-QA particularly challenging. First, answer spans are much longer than in general-domain QA benchmarks, averaging 35.6 words compared with 3.2 in SQuAD 1.1. As a result, Exact Match is a stricter metric, and accurate boundary prediction becomes more difficult. Second, the vocabulary is highly Act-specific. Terms such as *compulsory licence*, *geographical indication*, and *essentially derived variety* are largely absent from general pretraining corpora and are unevenly distributed across Acts. The Plant Varieties Act illustrates this challenge. Despite having the highest question–context lexical overlap (0.636), it achieves the lowest test F1 (63.27%). One possible explanation is its specialised biological terminology, which may create several plausible spans with similar lexical cues; this hypothesis is examined further in Section 7.2.

3.2 Seed Extraction and Selection

Legal entity seeds are extracted from the training corpus using two complementary methods. The first applies more than 60 rule-based legal patterns to identify IPR-specific terms such as *compulsory licence*, *patentee*, and *geographical indication*, weighted by legal specificity. The second extracts 1–3 gram keyphrases with KeyBERT (Grootendorst 2020) using all-MiniLM-L6-v2 (Reimers and Gurevych 2019), ranked by semantic relevance. After normalisation and stop-word filtering, the process yields 5,859 unique candidate seeds. These are ranked by frequency, filtered to a minimum count of three, and the top 1,500 are retained for passage generation.

The hybrid approach combines the strengths of both methods. Rule-based patterns capture stable statutory entities and recurring IPR terminology, while KeyBERT identifies additional corpus-derived phrases that are difficult to specify manually. The minimum frequency threshold of three filters out rare or fragmentary candidates, and frequency ranking provides a simple reliability signal before the computationally expensive generation stage. Retaining the top 1,500 seeds balances concept coverage with the CPU-only generation budget.

3.3 Synthetic Context Generation

Each seed generates passages in two modes. The first is the *solo* mode, which produces eight legal writing variants: definition-oriented, registration-oriented, rights and remedies, authority and procedure, conditions-and-compliance, penalty-and-enforcement, licensing-and-assignment, and dispute-and-appeal. The second is the *broad-supported* mode, which generates six additional passages by pairing the primary seed with a support seed from the same or a related Act. Together, these modes provide 14 generation slots per seed, yielding a target of approximately 21,000 passages per run.

Generation uses the Gemma 3 family (Gemma Team 2025), served locally through Ollama, resulting in 30,282 generation attempts across two runs. Gemma 3:12b is the primary generator and produces most of the accepted corpus, while a second pass with Gemma 3:27b increases phrasing diversity. The accepted outputs from both models are merged using exact-text deduplication. Per-generator statistics are reported in Table 4.

The two generation modes serve complementary purposes. Solo generation provides direct coverage of the primary legal concept under multiple statutory writing styles, whereas broad-supported generation introduces a second, related seed to model co-occurring legal concepts without encouraging arbitrary combinations. Support seeds are preferentially selected from the same Act and restricted to higher-frequency candidates. The style variants also include different target lengths to better reflect the variability of real statutory provisions. Gemma 3 was selected because it produced higher-quality legal text than the available alternatives within the CPU-only computation budget. Qwen 2.5:7b remains in the released pipeline as a lower-memory fallback but contributes no passages to the corpus used in this study.

3.4 Quality Filtering and Accepted Corpus

Every generated passage is validated using a rule-based pipeline. The checks include a length of 70–280 words, the presence of the primary seed entity, the presence of the support seed entity in broad mode, a valid sentence opening, no formatting artifacts (e.g., markdown or prompt leakage), at least one legal-style phrase (e.g., "shall" or "provided that"), at least two sentences, no textbook-style opening, and no excessive n-gram repetition, defined as the same 6-gram appearing three or more times. Each generation is allowed up to two retries before being rejected. Table 4 summarises the final generation results.

Table 4 Synthetic passage generation statistics. Accepted passages from the two generators are merged using exact-text deduplication.

Generator	Attempted	Accepted	Rejected	Acceptance rate
Gemma 3:12b	21,000	9,294	11,706	44.3%
Gemma 3:27b	9,282	3,917	5,365	42.2%
Combined	30,282	13,211	17,071	43.6%

Rejection reasons were aggregated across both generation runs, and a single rejected passage could fail multiple validation checks. The most common reason was a missing primary entity (13,844 cases), followed by a missing support entity (7,669 cases). Formatting-related errors were rare, with fewer than 25 cases for each type.

Table 5 summarises the archived contents of the three DAPT corpora. This audit is important because the archived authentic-only corpus contains 3,710 context records, one per training QA pair, that collapse to 901 distinct passages after normalisation, whereas the other corpora contain normalised unique or near-unique passages. Measured against the final split, 898 of those 901 distinct authentic passages are training contexts and three are test contexts; Section 6 audits how they entered and quantifies their effect. The mixed corpus (C') consists of 866 authentic records (865 unique after lowercase and whitespace normalisation), 305 pilot-generation passages, and 13,043 accepted passages from the main generation pool, giving 14,214 records and 14,213 normalised unique passages. The synthetic-only corpus (I') contains all 13,211 accepted main-generation passages.

The audit also checked for overlap between the DAPT corpora and the evaluation splits. No validation contexts appear in any DAPT corpus, and the synthetic-only corpus (I') contains no evaluation contexts. Three test contexts appear in the authentic-only (H') and mixed (C') corpora, corresponding to six of the 344 test questions (1.74%). These contexts are absent from the training split and therefore do not represent repeated statutory sections. Their exposure is limited to unsupervised masked language model (MLM) pretraining, with no associated questions or answer spans. Their impact is evaluated in Section 6.

Record multiplicity and corpus size differ across H', C', and I'. These configurations should therefore be interpreted as comparisons of complete training regimes rather than as a controlled comparison of authentic and synthetic text. Section 7.4 discusses the implications of these differences, including the checkpoint-selection asymmetry. The corpus files and a hash-based manifest are included in the anonymous review materials.

The DAPT stage introduces no new QA labels and exposes the model to no test answers. Instead, it familiarises the encoder with legal vocabulary and clause structure before supervised QA fine-tuning. For a passage $\mathbf{x} = (x_1, \ldots, x_n)$ and a masked token

Table 5 Composition of the archived DAPT corpora after auditing. Values in parentheses indicate the number of unique passages where this differs from the total record count.

Corpus	Authentic records	Pilot synthetic	Main synthetic	Total records
H' (Authentic-only)	3,710 (901)	0	0	3,710 (901)
C' (Mixed)	866 (865)	305	13,043	14,214 (14,213)
I' (Synthetic-only)	0	0	13,211	13,211

set $\mathcal{M}$, the MLM objective is

$$\mathcal{L}_{\text{MLM}} = -\frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \log P_\theta(x_t \mid \mathbf{x}_{\setminus \mathcal{M}}). \tag{1}$$

This objective is well suited to statutory QA because many errors arise from identifying the exact wording and boundaries of legal phrases rather than from high-level legal reasoning. By predicting masked legal terms in authentic and synthetic statutory contexts, the DAPT stage is intended to produce representations that may better support downstream answer-span extraction; Sections 5.4 and 5.5 test whether this expected benefit is detectable.

4 Experimental Setup

This section describes the experimental setup used to evaluate extractive QA over Indian IPR statutes. It presents the evaluation metrics, fine-tuning objective, model configurations, and training settings. All encoder experiments were conducted on a local workstation with an Intel(R) Core(TM) i9-14900 processor (24 cores, 32 logical threads, 2.0 GHz base clock) and 32 GB RAM. No GPU acceleration was used for training or evaluation. Although this CPU-only setup increases training time, it demonstrates that the experiments are reproducible on high-end commodity hardware.

4.1 Evaluation Metrics

Exact Match (EM) and token-level F1 are the primary evaluation metrics. Predictions and references are lowercased, stripped of punctuation, and normalised for whitespace before scoring. EM measures whether the normalised prediction exactly matches the reference:

$$\text{EM} = \mathbb{1}\{\text{norm}(\hat{a}) = \text{norm}(a)\}, \tag{2}$$

where a and $\hat{a}$ denote the gold and predicted answers after normalisation.

For token-level F1, let $\mathcal{A}$ and $\mathcal{P}$ denote the gold and predicted token multisets. Precision, recall, and F1 are computed as

$$\text{Precision} = \frac{|\mathcal{A} \cap \mathcal{P}|}{|\mathcal{P}|}, \qquad \text{Recall} = \frac{|\mathcal{A} \cap \mathcal{P}|}{|\mathcal{A}|}, \tag{3}$$

$$\mathrm{F1} = \frac{2\,\mathrm{Precision}\,\mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{4}$$

EM measures exact statutory span boundaries, whereas token-level F1 gives partial credit for overlapping spans. Character-frequency overlap is not reported because it ignores character order and may assign high scores to legally distinct spans.

4.2 Fine-Tuning Objective

The final task is extractive QA rather than text generation. The model predicts a contiguous answer span from the input statutory paragraph, ensuring that every answer is directly grounded in the source text without paraphrasing.

The encoder jointly represents the question and context, while two linear heads predict the start and end positions of the answer span. The predicted answer is the span with the highest joint start–end probability:

$$\hat{s},\, \hat{e} = \underset{i \le j}{\operatorname{argmax}}\; P(s{=}i \mid \mathbf{q}, \mathbf{c}) \cdot P(e{=}j \mid \mathbf{q}, \mathbf{c}) \tag{5}$$

$$\mathcal{L}_{\mathrm{QA}} = -\log P(s{=}s^*) - \log P(e{=}e^*). \tag{6}$$

The two prediction heads share the same contextual encoder and differ only in their task-specific linear projections. During training, the gold start and end positions (s^*, e^*) are obtained from the annotated answer spans. The loss in Eq. 6 therefore directly optimises span boundary prediction, which is particularly important for the long answer spans in IndicIPR-QA. Algorithm 1 summarises the complete training pipeline.

4.3 Model Inventory and Training Pipelines

Table 6 summarises the 11 evaluated model configurations. The experiments compare three training strategies: direct IPR fine-tuning, intermediate transfer through SQuAD v1, and three domain-adaptive pretraining (DAPT) variants built on InLegalBERT.

All four encoder families share the same transformer architecture: 12 self-attention layers, 768 hidden dimensions, 12 attention heads, and approximately 110 million parameters. They differ primarily in their pretraining data and objectives. BERT-base (Devlin et al. 2019) was pretrained on English Wikipedia and BookCorpus using masked language modelling and next-sentence prediction. LEGAL-BERT (Chalkidis et al. 2020) (`legal-bert-base-uncased`) was pretrained on 12 GB of Western legal text, including EU and UK legislation, European Court of Justice and European Court of Human Rights cases, US court decisions, and US commercial contracts. It is therefore a multi-jurisdictional legal encoder that does not include Indian legal text. InCaseLawBERT (Paul et al. 2023) was pretrained on Indian Supreme Court and High Court judgments, providing Indian legal vocabulary but a case-law-oriented representation space. InLegalBERT (Paul et al. 2023) was pretrained on Indian statutes, court judgments, and legal commentary, making it the closest existing encoder to the definition-heavy, clause-structured language of IndicIPR-QA.

Table 6 Evaluated model configurations and training sequences.

ID	Base model	Strategy	Training sequence
A	InLegalBERT	Direct	IPR
D	BERT-base	Direct	IPR
E	LEGAL-BERT	Direct	IPR
G	InCaseLawBERT	Direct	IPR
B'	InLegalBERT	Transfer	SQuAD v1 → IPR
$D2'$	BERT-base	Transfer	SQuAD v1 → IPR
$E2'$	LEGAL-BERT	Transfer	SQuAD v1 → IPR
F'	InCaseLawBERT	Transfer	SQuAD v1 → IPR
H'	InLegalBERT	DAPT	MLM (Authentic) → SQuAD v1 → IPR
C'	InLegalBERT	DAPT	MLM (Mixed) → SQuAD v1 → IPR
I'	InLegalBERT	DAPT	MLM (Synthetic) → SQuAD v1 → IPR

The four encoder families were selected to isolate the effects of domain, jurisdiction, and task format. BERT-base provides the general-domain baseline, LEGAL-BERT evaluates cross-jurisdictional transfer, InCaseLawBERT evaluates transfer from Indian case law to statutory QA, and InLegalBERT serves as the strongest Indian legal baseline and the foundation for DAPT. To ensure fair comparison, all direct and SQuAD-transfer models use the same final IPR fine-tuning configuration, so performance differences mainly reflect encoder initialisation and intermediate training history rather than optimisation settings. DAPT is applied only to InLegalBERT because it is the strongest baseline within the available CPU-only budget. SQuAD v1 is used for intermediate transfer because IndicIPR-QA contains only answerable questions. All transfer-based models use a fixed subset of 20,000 SQuAD v1 training examples and 2,000 validation examples.

4.4 Hyperparameters and Implementation Details

For DAPT, InLegalBERT-base is trained with a maximum input length of 256, a 15% masking rate, an effective batch size of 32, a learning rate of 2×10^{-5}, 100 warm-up steps, and 3 epochs. QA fine-tuning uses a maximum sequence length of 384, a document stride of 128, an effective batch size of 32 (batch size 8 with 4 gradient accumulation steps), and 3 epochs. The intermediate SQuAD v1 stage uses a learning rate of 3×10^{-5}, while the final IndicIPR-QA stage uses 2×10^{-5}.

Checkpoint selection differs slightly across model variants. The direct, SQuAD-transfer, and authentic-only DAPT (H′) models select the checkpoint with the highest validation F1, whereas the mixed (C′) and synthetic-only (I′) DAPT models select the checkpoint with the lowest validation loss, the default behaviour of the training framework. Because all models are trained for three epochs, this difference affects only checkpoint selection rather than the training schedule. Consequently, checkpoint selection is not a fully controlled factor and is discussed further in Section 7.4. Unless otherwise stated, all experiments use random seed 42, with one complete training run per configuration. The paired bootstrap intervals reported later therefore quantify variation across test examples rather than across training seeds. All experiments were conducted on the CPU-only workstation described in Section 4.

The learning rates and three-epoch schedule follow standard BERT fine-tuning practice (Devlin et al. 2019), while the 15% MLM masking rate follows the original BERT configuration and subsequent DAPT work (Gururangan et al. 2020). The DAPT input length is set to 256 because masked language modelling does not require the full question–context window used during QA fine-tuning.

For extractive QA, the input format is `[CLS] question [SEP] context [SEP]`. Long statutory contexts are split into overlapping windows with a stride of 128. Each window predicts start and end logits independently, and the final answer is selected using the highest joint confidence across all windows. Most IndicIPR-QA contexts fit within a single 384-token window because the average context length is 171.3 words, so multi-window inference is required only for the longest statutory passages.

5 Results

This section investigates the sources of the gap between gold-context and end-to-end performance. It first compares the 11 encoder configurations on the held-out test set and asks whether model adaptation accounts for that gap, analysing the effects of SQuAD transfer, DAPT, and MLM perplexity, and then controlling the apparent DAPT differences for corpus scale, composition, and training variability. It next examines split-wise and Act-level generalisation, retrieval performance, and question structure, together with comparisons against prompted open LLMs, before concluding with an error analysis, worked examples, and a dataset-quality ablation. The adaptation experiments produce only small numerical differences that this design cannot separate from training variability, whereas retrieval and the resolution of statutory references account for the dominant observed end-to-end failure on IndicIPR-QA.

5.1 Model Performance

The main comparison evaluates which training strategy produces the most reliable extractive QA model for Indian IPR statutes. All models are trained and evaluated on the same train, validation, and test splits, ensuring that differences reflect the training strategy rather than the data split or decoding configuration.

Table 7 shows that the mixed-corpus C′ pipeline achieves the highest test performance, with an F1 score of 82.33. Relative to direct InLegalBERT fine-tuning (A), C′ improves by 1.77 F1 points, and relative to the SQuAD-transfer baseline (B′), by 0.88 F1 points. Throughout the paper, performance gains are reported as absolute differences in F1:

$$\Delta_{\mathrm{F1}}(M, B) = \mathrm{F1}(M) - \mathrm{F1}(B) \tag{7}$$

where M denotes the evaluated model and B the baseline.

The direct fine-tuning baselines establish a clear ranking among the encoders. InLegalBERT performs best (80.56 F1), followed by LEGAL-BERT (79.75), InCaseLaw-BERT (77.88), and BERT-base (76.56). Legal-domain pretraining therefore improves performance, but jurisdiction and document genre also matter. The Indian legal

Table 7 Test performance on IndicIPR-QA. Model IDs correspond to Table 6. The best F1 score is shown in bold.

ID	EM (%)	F1 (%)
A	46.80	80.56
D	46.22	76.56
E	47.38	79.75
G	44.77	77.88
B′	47.38	81.45
D2′	47.97	78.68
E2′	47.38	81.10
F′	48.84	79.99
H′	47.97	81.43
I′	47.09	81.86
C′	48.26	**82.33**

encoder is the strongest under direct fine-tuning, whereas the case-law-oriented InCaseLawBERT is less effective for statutory span extraction. SQuAD v1 transfer improves all four encoder families, with gains of +0.89 F1 for InLegalBERT (A $\rightarrow$ B′), +1.35 for LEGAL-BERT (E $\rightarrow$ E2′), +2.11 for InCaseLawBERT (G $\rightarrow$ F′), and +2.12 for BERT-base (D $\rightarrow$ D2′). The larger gains for the weaker or more genre-mismatched encoders suggest that SQuAD transfer mainly teaches the general mechanics of extractive QA, while domain-specific pretraining remains important for Indian statutory language.

Exact Match scores are substantially lower than those typically reported on SQuAD, which is expected because IndicIPR-QA answers average 35.6 words, compared with about three words in SQuAD. Long statutory answers make exact span boundary prediction considerably more difficult, even when token overlap is high. Paired bootstrap resampling (10,000 iterations, seed 42) shows that C′ is numerically the strongest model, but its advantage is not statistically significant after Holm correction (Holm 1979). Compared with A, the F1 improvement is +1.77 (95% CI [−0.04, +3.56], Holm-adjusted $p = 0.218$). C′ also does not differ significantly from B′, H′, or I′ ($p = 0.690$ for all comparisons). These confidence intervals reflect variation across test questions only because the trained checkpoints remain fixed.

Training-seed variability was analysed separately by repeating only the final IndicIPR-QA fine-tuning stage for A and C′ under three random seeds (7, 42, and 99), while keeping the DAPT and SQuAD-transfer checkpoints unchanged. Table 8 summarises the results. Seed-to-seed variation is similar for both models: A ranges from 78.65 to 80.56 F1 (SD = 0.97), whereas C′ ranges from 80.26 to 82.33 (SD = 1.10). In both cases, the variation exceeds the performance gap between the models, and the score ranges overlap.

Despite this variability, C′ outperforms A under all three seeds, with gains of +1.77, +1.61, and +0.74 F1. The mean paired improvement is +1.37 F1 (SD = 0.55; paired $t = 4.32$, df = 2, $p = 0.050$). With two degrees of freedom this test establishes that the sign of the difference was consistent across the seeds tried, not the size of the effect, and it should be read that way. Because only three seeds were evaluated, the magnitude of the effect remains uncertain. The seed-averaged scores are 79.70 F1 for A and 81.08 F1 for C′, while the single-seed results reported elsewhere correspond to seed 42.

Table 8 Multi-seed repetition of the final IndicIPR-QA fine-tuning stage. The DAPT and SQuAD-transfer checkpoints are held fixed, so the variation reflects only the final fine-tuning stage. Seed 42 is used throughout the rest of the paper.

Seed	A: EM / F1	C′: EM / F1	ΔF1
42	46.80 / 80.56	48.26 / 82.33	+1.77
7	45.93 / 78.65	47.09 / 80.26	+1.61
99	47.38 / 79.90	46.80 / 80.64	+0.74
Mean	46.71 / 79.70	47.38 / 81.08	+1.37
SD	0.97	1.10	0.55

The bootstrap and multi-seed analyses provide complementary evidence. The bootstrap analysis shows that, on the 344-question test set, the seed-42 performance gap cannot be distinguished from question-sampling variability alone. In contrast, the multi-seed experiment shows that the same performance difference appears under all three seeds tested. The C′–A difference is therefore positive under every fine-tuning seed tried, but its magnitude remains uncertain, and C′ differs from A in its whole adaptation route rather than in synthetic DAPT alone. Accordingly, Table 7 should be interpreted as a cluster rather than a strict ranking. The four adapted InLegalBERT pipelines, B′, H′, I′, and C′, lie within a narrow 0.90-F1 band (81.43–82.33), and no pair differs significantly after Holm correction. Although all four outperform direct fine-tuning (A, 80.56 F1) numerically, even the largest improvement is not statistically significant. Adaptation beyond direct fine-tuning therefore produces modest numerical gains, but the experiment does not identify which adaptation component causes them: SQuAD transfer, authentic DAPT, synthetic DAPT, and mixed-corpus DAPT are not separable at the current test-set size ($n = 344$). The differing corpus sizes and record multiplicities of these strategies are discussed in Section 7.4.

5.2 Effect of SQuAD Transfer and Domain-Specific Adaptation

To test whether IndicIPR-QA can be solved through general extractive QA transfer alone, SQuAD-only models are evaluated directly on the test split using the gold statutory paragraph, without any IndicIPR-QA supervision. Table 9 compares these models with their corresponding IPR-supervised counterparts.

Table 9 SQuAD-only transfer diagnostic using gold statutory paragraphs. None of the models is fine-tuned on IndicIPR-QA. ΔF1 denotes the difference from the corresponding IPR-supervised model (B′, D2′, E2′, and F′ in Table 7).

SQuAD-only model	EM (%)	F1 (%)	F1 gap
InLegalBERT	3.20	25.57	−55.88
BERT-base	**6.98**	**34.85**	−43.83
LEGAL-BERT	3.78	24.93	−56.17
InCaseLawBERT	5.52	33.10	−46.89

The best SQuAD-only model reaches only 34.85 F1, well below the IPR-supervised models. This confirms that IndicIPR-QA cannot be solved through general extractive QA transfer alone. SQuAD transfer learns the mechanics of answer-span extraction but not the specialised terminology, statutory cross-references, or long-clause boundary conventions of Indian IPR law. After fine-tuning on IndicIPR-QA, all four encoders improve by 43.83–56.17 F1 points. The largest improvement is observed for LEGAL-BERT (56.17 F1 points), closely followed by InLegalBERT (55.88 F1 points), whereas BERT-base shows the smallest improvement (43.83 F1 points). Legal-domain pretraining alone is therefore insufficient for statutory span extraction. It becomes effective only when combined with in-domain supervision, which is the setting targeted by the DAPT pipeline evaluated here.

5.3 DAPT Perplexity and Downstream QA Performance

MLM perplexity on held-out statutory contexts is used as a diagnostic of domain fit. It is computed as:

$$\mathrm{PPL} = \exp\left(-\frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \log P\big(x_t \mid \mathbf{x}_{\setminus t}\big) \right), \tag{8}$$

where $\mathcal{M}$ denotes the set of masked positions. Table 10 shows that all three DAPT variants substantially reduce perplexity compared with the original InLegalBERT model.

Table 10 MLM loss and perplexity on held-out statutory contexts. Perplexity reduction is measured relative to the original InLegalBERT model.

Model	MLM loss	PPL	PPL reduction
InLegalBERT	2.4553	11.65	–
Authentic DAPT (H′)	0.9245	2.52	78.4%
Mixed DAPT (C′)	0.9511	2.59	77.8%
Synthetic DAPT (I′)	1.0337	2.81	75.9%

Authentic DAPT (H′) achieves the lowest perplexity (2.52) because it is trained only on authentic statutory text. However, the mixed DAPT model (C′) obtains the highest downstream QA performance, reaching 82.33 F1 compared with 81.43 for H′, despite its slightly higher perplexity. This shows that lower MLM perplexity does not necessarily translate into better extractive QA performance. The three DAPT variants differ not only in corpus composition but also in corpus size and record multiplicity, so the current experiments do not identify which of these factors drives the observed differences.

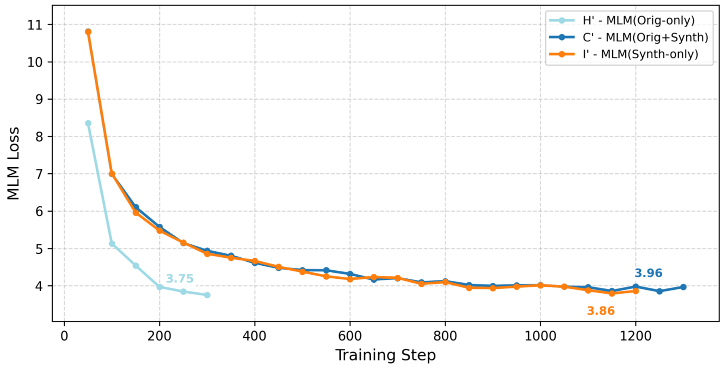

Fig. 2 MLM pretraining loss for H′, C′, and I′. H′ converges more quickly because it is trained on a smaller corpus.

Figure 2 shows stable optimisation for all three DAPT variants before QA fine-tuning. Section 5.4 separates the factors that these three variants confound.

5.4 Separating Corpus Composition from Corpus Scale

The three DAPT corpora in Table 5 differ in size, in record multiplicity, and in composition at the same time, so no difference among H', C', and I' can be attributed to a single factor. H', C', and I' should therefore not themselves be read as a causal comparison of corpus composition; the controlled composition analysis is the matched experiment reported in this section and in Section 5.5. This section reports a size-matched ablation designed to separate these factors as far as the archived corpora permit. The archived authentic pool holds 901 distinct paragraphs, so 901 is the largest size at which authentic and synthetic corpora can be matched without upsampling. Of these, 898 are the distinct normalised contexts of the training split and three are test contexts that entered when the corpus was built against an earlier state of the split, as audited in Section 6. Three corpora of exactly 901 records were therefore built: 901 distinct authentic paragraphs, 901 passages sampled from the accepted synthetic pool, and a mixture of 450 authentic and 451 synthetic passages. Each was run through the full pipeline of masked language model pretraining, SQuAD v1 transfer, and IPR fine-tuning with all hyperparameters unchanged. Every run in this section, including a re-trained H', retains the checkpoint with the lowest validation loss, so checkpoint selection is held fixed as well.

Table 11 Size-matched DAPT ablation. All three matched corpora hold exactly 901 records and every run uses the lowest-validation-loss checkpoint rule. Reference rows are the published variants, which differ in scale.

Corpus	Composition	Records	PPL	EM (%)	F1 (%)
Size-matched					
auth901	901 authentic	901	3.01	47.67	**81.92**
mixed901	450 auth. + 451 synth.	901	3.03	47.67	81.66
synth901	901 synthetic	901	3.18	**48.26**	81.19
Reference, not size-matched					
A	no DAPT	–	–	46.80	80.56
H'	901 authentic, repeated	3,710	2.49	47.97	81.00
I'	synthetic	13,211	2.79	47.09	81.86
C'	mixed	14,214	2.57	48.26	82.33

Perplexity is held-out MLM perplexity on the 113 distinct test contexts, recomputed for all rows in a single run; values differ slightly from Table 10 because masking is resampled. H' is listed here under the loss rule at 81.00 F1; Table 7 reports it under the F1 rule at 81.43.

At seed 42, Table 11 shows a small 0.73-F1 spread across the three size-matched corpora once scale is fixed. They span 81.19 to 81.92 F1, with authentic text highest and synthetic text lowest. Synthetic text therefore does not outperform authentic text at equal scale. Exact match runs in the opposite direction, with the synthetic corpus highest at 48.26, so the two metrics do not agree on an ordering.

Scale has an effect of similar size. Increasing the synthetic corpus from 901 to 13,211 passages raises F1 from 81.19 to 81.86, and increasing the mixed corpus from 901 to 14,214 passages raises it from 81.66 to 82.33. Both gains are 0.67 F1 for roughly fifteen times more pretraining text.

Record multiplicity acts in the opposite direction to expectation. H' trains on the same 901 distinct paragraphs as auth901 but repeats each about four times to give 3,710 records. Removing that repetition raises F1 from 81.00 to 81.92, so the repetition in the published authentic corpus was mildly harmful rather than helpful.

Held-out perplexity follows the same ordering as F1 inside the matched group, at 3.01 for authentic, 3.03 for mixed, and 3.18 for synthetic, but it does not predict F1 across groups. H' records the lowest perplexity of any variant at 2.49, yet auth901, at 3.01, scores higher on F1. This reinforces the earlier observation that MLM perplexity and downstream span F1 do not induce the same ranking.

Four quantities have now been measured with the others held fixed. Composition at fixed scale spans 0.73 F1, a fifteen-fold increase in scale is worth 0.67 F1, removing record multiplicity is worth 0.92 F1, and unifying the checkpoint rule moves one model by 0.43 F1. All four are smaller than the seed-to-seed standard deviation of about 1.0

F1 in Table 8, and two approach the run-to-run reproducibility of about 0.5 F1 noted in Section 5.17. Every adaptation configuration scores numerically above direct fine-tuning, from 80.56 to between 81.00 and 82.33 F1, but no single factor accounts for that difference at a magnitude this setup can resolve. These are single-seed runs, so the four values should be read as bounds on each effect rather than as estimates of it. Section 5.5 converts the first of them, composition, into an estimate.

5.5 Composition Under Three Seeds

Because every run in Table 11 uses seed 42, the 0.73 F1 composition spread bounds the effect without estimating it. To separate the effect from seed noise, the full pipeline—MLM pretraining, SQuAD v1 transfer, and IPR fine-tuning—was repeated end to end for each matched corpus under two further seeds, 7 and 99, giving three complete runs per corpus and nine in total. The seed is applied to all three stages rather than to fine-tuning alone, so the resulting spread covers DAPT-stage variance as well. Every run retains the lowest-validation-loss checkpoint, matching the seed-42 runs.

Table 12 Multi-seed repetition of the size-matched ablation. The full pipeline is repeated end to end under seeds 7, 42, and 99 for each matched corpus, so the spread covers DAPT-stage variance as well as fine-tuning variance. Cells are EM / F1; SD is over the three F1 values. The seed-42 column reproduces Table 11.

Corpus	Seed 7	Seed 42	Seed 99	Mean F1	SD
auth901	47.67 / 80.51	47.67 / 81.92	47.67 / 81.41	81.28	0.71
synth901	48.84 / 81.76	48.26 / 81.19	48.55 / 81.00	81.32	0.40
mixed901	46.51 / 80.81	47.67 / 81.66	48.84 / 81.62	81.36	0.48

No corpus is marked best: the spread between the three means is 0.08 F1 against a pooled within-corpus SD of 0.53 F1, and neither a one-way nor a seed-blocked analysis of variance detects a corpus effect ($p = 0.983$ and $p = 0.985$). Mean EM is 47.67 for auth901, 48.55 for synth901, and 47.67 for mixed901.

The composition effect does not survive averaging over seeds. The three corpus means lie within 0.08 F1 of one another, at 81.28 for authentic, 81.32 for synthetic, and 81.36 for mixed, while the pooled within-corpus standard deviation is 0.53 F1. Composition at fixed scale is therefore about seven times smaller than the seed noise it sits inside. It is now estimated rather than bounded, and the estimate is indistinguishable from zero.

The ordering is unstable as well. Under seed 42 the authentic corpus is highest and the synthetic corpus lowest, the 0.73 F1 spread reported in Table 11. Averaged over three seeds that ordering reverses, with mixed nominally highest and authentic nominally lowest, and the three runs of a single corpus span up to 1.41 F1—wider than the gap between any two corpus means. The seed-42 ordering is therefore not reproducible across the three seeds tested. Exact match again disagrees with F1, placing the synthetic corpus highest at 48.55 against 47.67 for the other two; with 344 test questions one item is worth 0.29 EM points, so that difference is three questions.

A one-way analysis of variance over the nine runs finds no detectable corpus effect on F1, $F(2, 6) = 0.02$, $p = 0.983$, $\eta^2 = 0.006$. Because the same three seeds were used for every corpus, seed is a genuine blocking factor, and a randomised-block analysis is the more appropriate test; it agrees, $F(2, 4) = 0.02$, $p = 0.985$, with the block itself also not significant ($F = 0.73$, $p = 0.538$). Composition thus accounts for well under one percent of the variance among the nine runs, and what variance there is belongs neither to the corpus nor to the seed but to the residual.

The power of the design must be stated alongside that result. Paired differences over the three shared seeds are -0.03 F1 for authentic against synthetic, -0.08 for authentic against mixed, and -0.05 for synthetic against mixed, all far below their standard errors, but with $n = 3$ the corresponding 95% confidence intervals are wide: $[-2.67, +2.60]$, $[-0.84, +0.67]$, and $[-2.19, +2.10]$ respectively. The experiment therefore excludes a composition effect larger than about 2.7 F1, which is wider than the $+1.77$ F1 that separates C′ from direct fine-tuning. This is a non-detection with a weak bound, not a demonstration that the effect is exactly zero. What it does establish is that the 0.73 F1 ordering observed at seed 42 is not reproducible, and that the experiment provides no detectable evidence that corpus composition explains the adaptation gain at the resolution of the present test set and seed budget.

Exact match requires one caution. The authentic–synthetic contrast is nominally significant before correction ($t = -5.20$, $p = 0.035$), with the synthetic corpus ahead, but the comparison does not survive Holm correction across the three pairs ($p = 0.105$). It is also fragile: auth901 returns identical exact match at all three seeds, so the paired standard deviation is supplied entirely by synth901, and the gap is three questions out of 344. It should not be read as evidence that synthetic text helps.

The pooled standard deviation of 0.53 F1 is smaller than the 0.97–1.10 in Table 8, even though it varies the seed across all three stages rather than the final stage alone. The two are measured on different systems—Table 8 compares A and C′, which differ in pretraining scale by more than an order of magnitude—so they are not competing estimates of one quantity. Read together, they place seed noise for this pipeline between roughly 0.5 and 1.1 F1, which exceeds all four of the effects measured in Section 5.4. Scale, record multiplicity, and the checkpoint rule remain single-seed bounds; only composition has been repeated.

5.6 Split-wise Generalisation

The train–test F1 gap is used as a diagnostic of generalisation:

$$\Delta_{\text{gen}}(M) = \text{F1}_{\text{train}}(M) - \text{F1}_{\text{test}}(M). \tag{9}$$

Smaller values indicate that test performance is closer to training performance, although they do not by themselves imply better generalisation.

Table 13 shows that C′ achieves the highest validation and test F1 scores while maintaining a train–test gap of 4.93 points. Its training F1 is nearly identical to that of B′, H′, and I′ (87.25–87.35), indicating that its higher test performance is not simply

Table 13 Split-wise F1 scores and train–test gap for all models. Test F1 values match Table 7; training and validation scores are diagnostic evaluations of the same retained checkpoints. Best value per column is shown in bold.

Model	Train F1	Val F1	Test F1	Gap
A	84.39	78.31	80.56	3.83
D	83.45	72.64	76.56	6.89
E	84.17	75.54	79.75	4.42
G	80.87	69.93	77.88	**2.99**
B′	**87.35**	83.13	81.45	5.90
D2′	84.50	78.91	78.68	5.82
E2′	86.46	81.22	81.10	5.36
F′	86.26	78.42	79.99	6.27
H′	87.29	82.63	81.43	5.86
I′	87.25	83.06	81.86	5.39
C′	87.26	**83.69**	**82.33**	4.93

the result of fitting the training data more closely. Although G has the smallest gap (2.99), it also has a substantially lower test F1 (77.88), showing that a small train–test gap alone is not evidence of better generalisation. Because IndicIPR-QA uses section-level splitting, the test set contains statute sections unseen during training, making this analysis a useful diagnostic. The official performance comparison, however, remains the held-out test results in Table 7.

5.7 Per-Act Performance and Difficulty Analysis

Table 14 reports per-Act performance for the best model, C′. Performance varies substantially across Acts. The Designs and Patents Acts are the easiest, with F1 scores of 99.42 and 97.54, whereas the Plant Varieties Act is the hardest at 63.27 F1. These results should be interpreted exploratorily because the test split contains only 24 questions for the Designs Act and 88 for the Trade Marks Act, giving wide sampling uncertainty for individual Act scores. The correlation analysis below is also based on only seven observations. The ordering is reported because it is consistent across all 11 models rather than because any individual per-Act estimate is highly precise.

The same difficulty ordering—Designs, Patents, Semiconductor, GI, Copyright, Trade Marks, and Plant Varieties—is observed across all 11 models (Fig. 3), suggesting that the differences mainly reflect the statutory text rather than a particular model. Table 15 reports exploratory Pearson correlations between selected linguistic features and per-Act F1.

Question–context lexical overlap shows the strongest association with Act-level F1 ($r = -0.785$). This association is not robust: the leave-one-Act-out check in Section 5.9 shows it ranging from -0.927 to -0.549, falling to -0.549 once the Plant Varieties Act is removed, so it should be treated as a description of these seven Acts rather than as a finding. The Plant Varieties Act illustrates the pattern: despite its high lexical overlap, it has the lowest F1. One possible explanation is the presence of specialised biological

Table 14 Per-Act test performance of the best model (C′).

Act	Test records	EM (%)	F1 (%)
Designs Act	24	87.50	99.42
Patents Act	69	65.22	97.54
Semiconductor Act	47	34.04	88.59
GI Act	51	56.86	82.33
Copyright Act	25	48.00	80.90
Trade Marks Act	88	34.09	71.45
Plant Varieties Act	40	32.50	63.27
Overall	344	48.26	82.33

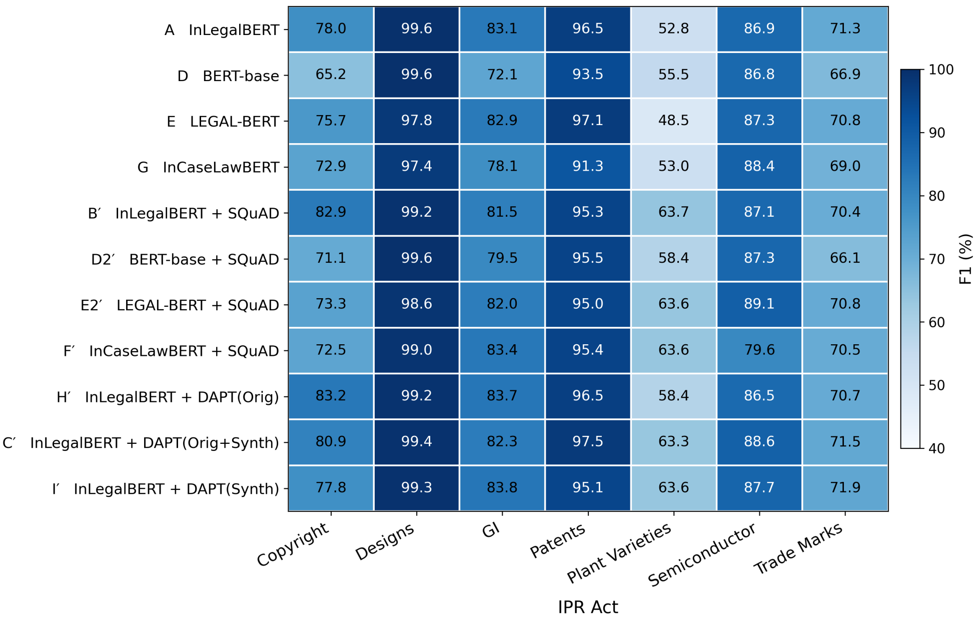

Fig. 3 Per-Act F1 heatmap for all 11 models. The same difficulty ordering is observed across all training pipelines.

Table 15 Exploratory Pearson correlations between selected Act-level features and test F1 for C′ ($n = 7$).

Feature	r with Test F1
Q–context lexical overlap	−0.785
Average answer position	−0.441
Multi-clause answer rate (%)	+0.471
Average answer length (words)	+0.247
Answer vocabulary TTR (%)	+0.226
Average context density	−0.029

terms such as "essentially derived variety" and "propagating material", which may be poorly represented in general and legal pretraining corpora. Average answer position also correlates negatively with F1 ($r = -0.441$), suggesting that answers appearing later in a paragraph are harder to extract. The positive correlations for answer length and multi-clause rate should not be interpreted causally because the analysis is based on only seven Acts. They mainly reflect that the Designs and Patents Acts contain long but clearly delimited answers, whereas the Plant Varieties Act contains shorter answers with more specialised terminology.

5.8 Retrieval-Based Evaluation

The previous experiments evaluate the reader with the gold statutory paragraph provided. The end-to-end setting instead requires BM25 to retrieve candidate contexts before C′ extracts the answer span. For a question q and candidate context c, BM25 scores lexical relevance as:

$$\mathrm{BM25}(q, c) = \sum_{t \in q} \mathrm{IDF}(t) \, \frac{f(t, c)(k_1 + 1)}{f(t, c) + k_1 \left(1 - b + b\frac{|c|}{\overline{|c|}}\right)}, \tag{10}$$

where $f(t, c)$ is the frequency of query term t in context c, $|c|$ is the context length, and $\overline{|c|}$ is the average context length. BM25 is a natural baseline because it rewards exact lexical matches, but this also limits its effectiveness on statutory text, where many paragraphs share similar legal terminology while describing different rights, exceptions, or procedures. In the end-to-end pipeline, BM25 retrieves from the 1,116 statutory contexts in the curated corpus. C′ then extracts an answer span from each retrieved context, and the highest-confidence prediction among the top-k candidates is returned:

$$\hat{c} = \underset{c \in \mathrm{Top}_k(q)}{\operatorname{argmax}} \; S_\theta(q, c), \tag{11}$$

$$\hat{a} = a_\theta(q, \hat{c}). \tag{12}$$

Here $\mathrm{Top}_k(q)$ denotes the BM25-ranked candidate set, $S_\theta(q, c)$ is the reader's confidence score, and a_θ is the predicted answer span.

Table 16 End-to-end retrieval and extractive QA using C′ on IndicIPR-QA.

Setting	R@k (%)	EM (%)	F1 (%)
Gold paragraph + C′	100.00	48.26	82.33
BM25 top-1 + C′	13.66	8.72	24.08
BM25 top-3 + C′	20.06	9.88	29.22
BM25 top-5 + C′	22.97	9.88	30.00

Table 16 shows that retrieval is the main bottleneck in the end-to-end pipeline. With the correct paragraph supplied, C′ achieves 82.33 F1. When BM25 retrieves from the full corpus, recall reaches only 13.66% at top-1 and 22.97% at top-5. The reader therefore often extracts a plausible legal span from an incorrect paragraph. Increasing the retrieval depth from top-1 to top-5 improves F1 from 24.08 to 30.00, but end-to-end performance remains far below the gold-paragraph setting. These results indicate that further improvements are more likely to come from stronger legal retrieval or re-ranking than from additional reader optimisation.

5.9 Can Off-the-Shelf Dense Retrieval Remove the Bottleneck?

The previous experiment shows that BM25 retrieval is the main bottleneck in the end-to-end pipeline, but it does not establish whether this limitation is specific to BM25 or reflects the difficulty of statutory retrieval itself. To answer that question, four off-the-shelf dense retrievers and a sparse–dense hybrid are evaluated on the same corpus of 1,116 statutory passages and the same 344 test questions. This retrieval-only experiment excludes the reader and therefore isolates candidate selection. The dense retrievers encode questions and passages independently and rank them by cosine similarity, while the hybrid combines BM25 with the strongest dense retriever using reciprocal rank fusion (k=60).

Table 17 Retrieval-only comparison on 1,116 statutory passages and 344 test queries. Best value in each column is shown in bold.

Retriever	R@1	R@3	R@5	R@10	MRR@10
BM25 (sparse)	**13.08**	**20.06**	22.67	27.03	17.34
all-MiniLM-L6-v2	10.17	16.28	20.93	28.20	14.70
multi-qa-MiniLM-L6-cos-v1	8.43	16.28	17.73	24.13	12.66
bge-small-en-v1.5	10.17	18.31	22.38	30.23	15.80
e5-small-v2	9.88	18.31	23.26	29.94	15.60
Hybrid RRF (BM25 + e5-small)	11.92	19.77	**26.16**	**31.40**	**17.47**

All rows come from a single retrieval-only pipeline, so comparisons within this table are exact. Its BM25 is an independent reimplementation of the retriever used in Table 16 (same tokenizer, k_1=1.5, b=0.75) and reproduces that table's recall to within 0.6 points (13.08/20.06/22.67 here against 13.66/20.06/22.97 there at k=1/3/5). The residual difference comes from tie-breaking among equally scored passages and does not affect any comparison drawn here.

Table 17 shows that no dense retriever consistently outperforms BM25. The strongest dense model, e5-small-v2, improves recall@5 only slightly (23.26 versus 22.67) but performs worse at rank 1 (9.88 versus 13.08) and on MRR. Sparse and dense retrieval are partly complementary, however. Their hybrid increases recall@5 to 26.16 and recall@10 to 31.40, the best results obtained. Even so, the best configuration retrieves the governing paragraph within the top five for only about one question in four. Zero-shot dense retrieval therefore does not remove the retrieval bottleneck, although this experiment does not rule out dense retrievers trained on legal or in-domain question–paragraph pairs.

Table 18 Per-Act recall@5, lexical overlap, and gold-context F1. Retrieval performance and gold-context extraction follow opposite trends. The final retrieval column adds cross-encoder re-ranking (Section 5.11).

Act	Q–Ctx overlap	BM25 R@5	Hybrid R@5	Hybrid+CE R@5	Reader F1
Plant Varieties	0.636	45.00	50.00	72.50	63.27
Copyright	0.447	32.00	36.00	36.00	80.90
GI	0.458	25.49	31.37	35.29	82.33
Patents	0.429	24.64	23.19	26.09	97.54
Trade Marks	0.416	17.05	22.73	21.59	71.45
Semiconductor	0.418	14.89	19.15	23.40	88.59
Designs	0.320	0.00	0.00	0.00	99.42

The per-Act results in Table 18 reveal a pattern hidden by the overall averages. Question–context lexical overlap, which Table 15 identified as the strongest negative correlate of reader F1 ($r{=}-0.785$), shows the opposite relationship with retrieval recall ($r{=}+0.93$ for both BM25 and the hybrid, $n{=}7$). Reader F1 and retrieval recall are therefore strongly anti-correlated across Acts ($r{=}-0.789$ against hybrid recall@5). In other words, the same property that makes retrieval easier tends to make answer extraction harder. Figure 4 illustrates this retrieval–reading tension.

Because these correlations rest on seven points, each was recomputed with one Act removed at a time. The association between lexical overlap and retrieval recall is stable, remaining within $[+0.898, +0.969]$ for BM25 and $[+0.915, +0.971]$ for the hybrid, and it is weakest when the Designs Act is dropped. The association between lexical overlap and reader F1 is not stable. It ranges from -0.927 to -0.549 and falls to -0.549 once the Plant Varieties Act is removed. The retrieval side of the tension should therefore be read as a robust pattern, and the reading side as a suggestive one that a single Act can weaken.

The Designs Act illustrates this trade-off most clearly. It has the highest reader F1 in the benchmark (99.42) despite the lowest lexical overlap (0.320), yet neither BM25 nor the hybrid retrieves any of its 24 gold paragraphs within the top five. Many Designs questions are specified mainly by structural references rather than by content, for example "Under Section 7, what does the section say?" or "What is set out in sub-section (2) of Section 8?" Once the correct paragraph is retrieved, these questions are easy for the reader because the answer is essentially the cited clause. However, they provide few content terms for retrieval, since words such as *section*, *sub-section*, and *clause* recur throughout the statutes. Cross-Act textual similarity makes the problem harder. For example, one Designs paragraph has a token Jaccard similarity of 0.95 with a Patents Act paragraph, making lexical or semantic matching alone insufficient to distinguish them.

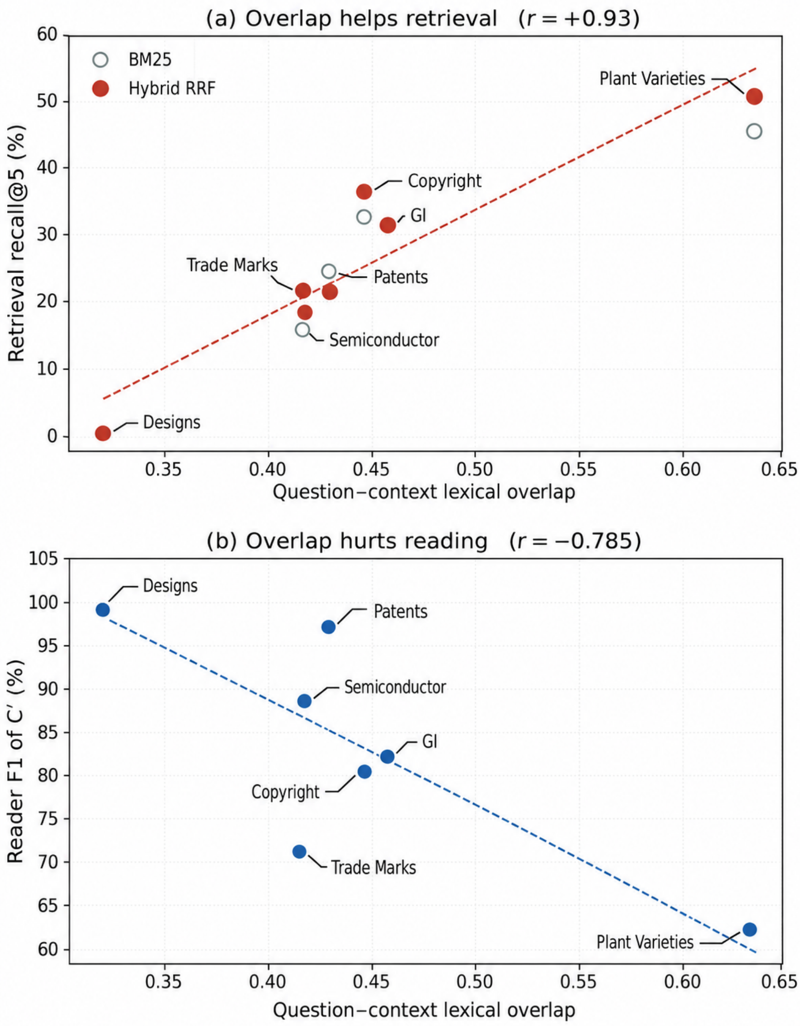

Fig. 4 Retrieval–reading tension across the seven Acts. Lexical overlap improves retrieval recall@5 (a) but reduces reader F1 (b). Dashed lines are least-squares fits; correlations are exploratory (n=7). The Designs Act illustrates the strongest trade-off, with the lowest lexical overlap, the highest reader F1 (99.42), and no gold paragraph retrieved within the top five.

5.10 Citation-Specified Questions Distort Benchmark Difficulty

The Designs Act is not an isolated case. Applying a simple test to every test question, namely removing the structural citation and the stock interrogative frame and then checking whether any content word remains, classifies 177 of the 344 test questions (51.5%) as citation-specified. Table 19 separates the two groups. Citation-specified questions are considerably easier to extract from once the paragraph is supplied, at 88.63 F1 against 75.65 for content-bearing questions, and close to unretrievable, at

16.80 end-to-end F1 and 0.00% exact match against 43.99 F1 and 20.36% exact match. Their share also varies sharply across Acts, from 95.8% for Designs to 0.0% for Plant Varieties, which explains why those two Acts sit at opposite ends of both the reading and the retrieval rankings. The headline reader score of 82.33 F1 is therefore an average over an easier half of the benchmark and a harder half, and the content-bearing subset at 75.65 F1 is the more conservative estimate of reading performance on questions that identify their provision by content.

Table 19 Citation-specified versus content-bearing test questions, scored with C′.

		Gold paragraph		BM25 top-5	
Question type	n	EM (%)	F1 (%)	EM (%)	F1 (%)
Citation-specified	177	55.93	88.63	0.00	16.80
Content-bearing	167	40.12	75.65	20.36	43.99
All	344	48.26	82.33	9.88	30.00

A question is counted as citation-specified when no content word remains after removing its structural citation (Section, sub-section, clause, or item numbers) and the stock interrogative frame. The per-Act share ranges from 95.8% (Designs) to 0.0% (Plant Varieties).

The variation across Acts bears on how far this finding travels. IndicIPR-QA questions were machine-generated, so a natural objection is that the citation-specified share reflects the generation procedure rather than anything about statutes. The per-Act distribution argues against a uniform generator artifact. Every Act was processed by the same generator under the same prompt and the same filtering, yet the share spans almost the full range: 95.8% for Designs (23 of 24), 71.0% for Patents, 64.0% for Copyright, 60.2% for Trade Marks, 51.1% for Semiconductor Layout-Design, 23.5% for Geographical Indications, and 0.0% for Plant Varieties (0 of 40). A uniform procedure-level artifact would be expected to apply more consistently across Acts.

What varies is the text being queried. A provision that is defined largely by cross-reference offers little content to ask about, so a question targeting it tends to reduce to its citation, whoever or whatever writes the question. The retrieval consequence follows the same gradient rather than merely correlating with it. Ordering the seven Acts by citation-specified share reproduces the reverse ordering of their re-ranking gain exactly, without a single inversion: Plant Varieties gains +27.50 recall@5 at 0.0% citation-specified, then Geographical Indications +9.80, Semiconductor Layout-Design +8.51, Trade Marks +4.54, Copyright +4.00, Patents +1.45, and Designs +0.00 at 95.8% (Spearman $\rho = -1.00$, $p < 0.001$; Pearson $r = -0.92$). Re-ranking gain is the quantity the mechanism predicts, because a re-ranker can only reorder what first-stage retrieval already returned: it helps in proportion to the content a question offers, and cannot help at all where the gold paragraph never enters the candidate pool, as for Designs (Section 5.11). The variation is therefore consistent with an interaction between statutory drafting structure and the information available to the question generator, rather than with a uniform benchmark-generation artifact. Whether the same bottleneck appears elsewhere is left as an empirical question rather than a prediction, and it is one with a cheap answer: the classifier used here operates on question text alone, so the citation-specified share of any question set can be measured directly, before any retrieval system is built. That quantity, not the present result, is what determines whether these findings transfer. With seven Acts the correlation reported above remains an exploratory observation, consistent with the treatment of the per-Act correlations in Section 5.7, and it does not substitute for replication on an independently constructed corpus.

The content-bearing group is not equivalent to the citation-free group. Of the 167 content-bearing questions, 148 contain both a statutory citation and substantive content, while 19 contain no parsable citation at all. Citation presence and content-bearingness are therefore overlapping rather than mutually exclusive properties, and the two decompositions used in this paper—177 citation-specified against 167 content-bearing, and 325 citation-carrying against 19 citation-free—should not be read interchangeably.

Two implications follow. For benchmark design, a citation-specified question primarily tests citation-conditioned span extraction: it hands the reader the location of the answer and asks it to reproduce the clause. Roughly half of IndicIPR-QA therefore measures citation lookup rather than content-based identification, and the aggregate 82.33 F1 averages a citation-driven subset with a content-bearing one. The content-bearing figure of 75.65 F1 is the more informative estimate of reading ability. Because gold-context and retrieved-context evaluation measure different capabilities on these items, they should be reported separately, as done here.

This motivates a concrete revision. A second version of IndicIPR-QA should require every question to identify its provision by content, so that a reader cannot succeed by copying a clause whose location the question already supplied and a retriever is not asked to match on terms that recur across all seven Acts. Where a citation is retained it should accompany substantive content rather than replace it. Until such a revision exists, results on this benchmark, and on any statutory QA benchmark built the same way, should be reported separately for citation-specified and content-bearing items, because a single aggregate hides the fact that the two measure different things.

For citation-heavy queries in this benchmark, the results point first to resolving structural references rather than to further strengthening the semantic retrieval approaches tested here. Section 5.12 tests that directly.

5.11 Does Cross-Encoder Re-ranking Help?

The retrievers compared so far all score a passage without reference to the question, which limits how much question-passage interaction they can capture. A cross-encoder instead encodes the question and the passage jointly, allowing token-level interaction at a much higher computational cost. Scoring the whole corpus this way is impractical, so the cross-encoder is applied as a second stage. BM25 and the sparse–dense hybrid each retrieve 50 candidates, and `ms-marco-MiniLM-L-6-v2` re-orders them (Nogueira and

Cho 2019). The corpus, the 344 test questions, and the gold matching procedure are unchanged. Because re-ranking can only reorder what the first stage returns, recall@50 of the first stage is a hard ceiling and is reported alongside.

Table 20 Cross-encoder re-ranking of the top 50 first-stage candidates. Recall@50 is the first-stage ceiling that re-ranking cannot exceed. Best value per column in bold.

System	R@1	R@3	R@5	R@10	R@50	MRR@10
BM25	13.08	20.06	22.67	27.03	**41.86**	17.34
+ cross-encoder	**17.44**	**26.45**	29.94	34.59	**41.86**	**22.56**
Hybrid RRF	11.92	22.09	26.74	33.72	40.41	18.12
+ cross-encoder	17.15	24.71	**30.23**	**35.17**	40.41	22.30

The hybrid rows differ slightly from Table 17 because fusion is computed over depth-50 rather than depth-10 candidate lists; all other settings are identical.

Table 20 shows that cross-encoder re-ranking is the first intervention in this study that clearly improves retrieval. Applied to BM25, it raises recall@5 from 22.67 to 29.94 and recall@1 from 13.08 to 17.44. Applied to the hybrid, it reaches 30.23 recall@5, the best value obtained anywhere in these experiments and 4.07 points above the strongest first-stage system. The gain from re-ranking alone is therefore larger than the combined gain from four dense retrievers and reciprocal rank fusion.

Two limits are equally clear. First, the first stage bounds everything that follows. Recall@50 is 41.86 for BM25 and 40.41 for the hybrid, so even a perfect re-ranker could not place the governing paragraph in the top five for more than about two questions in five. Re-ranking already recovers a substantial part of that headroom, which means further progress depends on first-stage recall rather than on a stronger re-ranker. Second, even after re-ranking, the governing paragraph is still missing from the top five for roughly seven questions in ten.

The per-Act pattern supports the diagnosis of Section 5.9. The Plant Varieties Act, the only Act with no citation-specified questions, improves the most, from 50.00 to 72.50 recall@5. The Designs Act, which is 95.8% citation-specified, remains at 0.00 because its gold paragraphs are absent from the top 50 altogether, so no re-ordering can recover them. Re-ranking therefore helps where questions carry content and does nothing where they do not.

5.12 Citation-Aware Retrieval

The preceding sections argue that the residual retrieval failures are structural. This section tests that argument rather than asserting it. Each passage carries a title of the form “Section n: . . . ”, from which a section identifier is parsed by regular expression; the same expression is applied to the question text. No gold relevance labels or gold passage identities are used; section identifiers are parsed automatically from passage titles and question text. Questions cite a section without naming an Act, so a cited number still occurs in several Acts and the retriever must still disambiguate. A citation is parsed for 94.48% of test questions. Two variants are evaluated: a boost that adds a large constant to the score of every passage whose section matches, and a filter that restricts candidates to matching passages and falls back to unrestricted BM25 when nothing matches. Because the boost is large enough to dominate, the two are indistinguishable through rank 5 (50.29, 73.26 and 86.34) and diverge only in the tail, where the boost reaches 95.64 recall@10 against 95.35 for the filter. The boost variant is reported throughout. The filter falls back to unrestricted BM25 for exactly the 19 questions carrying no parsable citation.

Table 21 Citation-aware retrieval against the learned alternatives, on the same 1,116 passages and 344 questions. The section index contains no learned component.

System	R@1	R@3	R@5	R@10	MRR@10
BM25	13.08	20.06	22.67	27.03	17.34
Hybrid RRF (BM25 + e5-small)	11.92	19.77	26.16	31.40	17.47
BM25 + cross-encoder	17.44	26.45	29.94	34.59	22.56
BM25 + section index	**50.29**	**73.26**	**86.34**	**95.64**	**64.84**

The hybrid row reproduces Table 17, which fuses depth-10 candidate lists; Table 20 reports the same system over depth-50 lists, where fusion gives slightly different values. The BM25 and cross-encoder rows are identical across all three tables.

The index does not trivially resolve the question. Every passage title yields a section identifier, and a cited section narrows the corpus from 1,116 passages to 7.89 on average, with a median of 8 and a maximum of 15. Those candidates span 4.69 distinct Acts on average, and for 301 of the 325 matched queries, or 92.6%, the cited number occurs in more than one Act. Because questions name a section but not an Act, lexical scoring must still choose the governing statute from several candidates carrying the same section number. That is why recall@1 reaches 50.29 rather than approaching 100, and it is the residual task that a citation-aware system still has to solve.

Table 21 and Figure 5 show that indexing statutory structure is far more effective than any learned retriever tested. Recall@5 rises from 22.67 to 86.34 and recall@10 to 95.64. The re-ranker of Section 5.11 was bounded by a first-stage recall@50 of about 42%; the section index is not, because it changes candidate generation rather than re-ordering an existing list. The per-Act gains are largest exactly where the earlier systems failed worst: the Designs Act rises from 0.00 to 58.33 recall@5, Trade Marks from 17.05 to 94.32, and Semiconductors from 14.89 to 87.23. By question type, citation-specified items rise from 0.00 to 83.62 and content-bearing items from 46.71 to 89.22.

The decomposition by whether a question contains a citation is the more revealing result. For the 325 questions that cite a section, plain BM25 reaches 20.31 recall@5 and the section index reaches 87.69. For the 19 that do not, both score 63.16, since there is no citation to exploit. In this test set, plain BM25 therefore reaches roughly three times the recall@5 on citation-free questions as on citation-carrying questions, 21_retrieval_progression.pdf

Fig. 5 Retrieval progression on the 344 test questions. *(a)* Recall@5 for every system evaluated. The five non-index systems sit below the first-stage recall@50 ceiling that bounds any re-ranker; the section index is not bounded by it, because it changes candidate generation rather than re-ordering. *(b)* The same comparison split by question type. Citation-specified questions move from 0.00 to 83.62 and account for most of the overall improvement, roughly two-thirds of the additional top-5 retrieval successes.

although the citation-free estimate rests on only 19 items. A substantial part of the unusually low retrieval performance on this benchmark is attributable to how its questions are written; this does not imply that statutory passages are intrinsically easy to retrieve.

Two limits should be kept in view. The citation-free subset has only 19 questions, so the 63.16 figure is imprecise. And the method works because IndicIPR-QA questions almost always carry a citation; a deployment in which users describe a problem in their own words would not benefit in the same way, and there the content-bearing behaviour of the underlying retriever is what matters.

5.13 Intervening on the Statutory Reference

The preceding result invites an obvious objection: the section index succeeds because nearly every question literally contains a section number, so its advantage may be an artefact of the query format rather than evidence about retrieval. This section tests that by intervention rather than argument. To decompose the contribution of statutory-reference information and of substantive content, four query variants are constructed: the question as released (*original*); the citation deleted with no other edit (*citation removed*); the citation alone, with the substantive question deleted (*citation only*); and the section number replaced by a different section *of the same Act*, every other word unchanged (*wrong citation*). The citation-removed and wrong-citation conditions preserve the substantive wording; the citation-only condition deliberately discards it in order to isolate the reference signal. Deletions use the rule that defines the citation-specified split in Section 5.10, so the intervention and the classification are the same operation. Corrupted references are drawn from within the gold Act so that the replacement remains structurally plausible; the gold Act is used only to sample that replacement and is never supplied to any retriever. The condition is repeated under three corruption seeds. Questions carrying no parsable citation are never given a synthetic variant; they are reported separately as a natural control. Of the 344 test questions, 322 carry exactly one citation, three carry two or more, and 19 carry none, so all 325 citation-carrying questions are eligible under every condition. Where a question carries more than one citation, the deletion rule removes every reference, while the citation-only and wrong-citation conditions operate on the first; with three such questions this choice cannot affect the conclusions. Resolution is section-level throughout; sub-section and clause structure are not modelled. No retriever is retrained.

Table 22 reports recall@5 for every condition. Two hypotheses were declared before the experiment was run, both concerning citation-aware retrieval on the citation-specified questions:

H1 Deleting the statutory citation reduces recall@5.
H2 Replacing the citation with another valid section of the same Act reduces recall@5.

Both were strongly supported. Deleting the citation moves recall@5 from 83.62 to 1.13, a drop of 82.49 points (McNemar, $p = 7.4 \times 10^{-43}$). Replacing the section number, changing nothing else, moves it from 83.62 to 0.00, a drop of 83.62 points ($p = 1.9 \times 10^{-43}$), and the collapse is identical under all three corruption seeds (0.00 ± 0.00). All p-values reported in this section are Holm-corrected over the full family of 36 paired contrasts (six systems, three groups, two contrasts each), not over the two pre-specified hypotheses alone.

The wrong-citation result should be read as two distinct findings. As mechanism validation it is direct: the citation-aware gain is mediated by the parsed reference, since editing one number while leaving every content word in place removes the entire advantage. It is not surprising in isolation, because the system deliberately privileges

Table 22 Citation contribution ablation, recall@5, for representative systems; the full six-system results are released with the code. The citation-removed and wrong-citation conditions preserve substantive query content, while the citation-only condition isolates the statutory reference. The wrong-citation column replaces the section number with another section of the same Act; the citation-free control receives no synthetic variant.

Group	System	Original	Cit. removed	Cit. only	Wrong cit.
Citation-specified ($n = 177$)					
	BM25	0.00	1.13	0.00	0.56
	Hybrid + CE	0.56	0.00	0.00	1.69
	Citation-aware	**83.62**	1.13	62.71	**0.00**
Content-bearing, citation-carrying ($n = 148$)					
	BM25	44.59	42.57	2.03	43.92
	Hybrid + CE	60.14	54.05	10.14	49.32
	Citation-aware	**92.57**	42.57	65.54	11.49
Citation-free control ($n = 19$)					
	BM25	63.16	63.16	—	—
	Citation-aware	63.16	63.16	—	—

Wrong-citation recall@5 over three corruption seeds: 0.00 ± 0.00 for citation-aware on citation-specified questions and 12.61 ± 1.03 on content-bearing questions.

passages matching the parsed section, and a system told the wrong section will follow it. The second finding is operational, and less obvious: an erroneous reference is worse than no reference at all. Under corruption citation-aware retrieval reaches 11.49 recall@5 while plain BM25 on the same questions reaches 43.92, so a system that trusts a supplied citation without checking it against content can be driven below the lexical baseline it was meant to improve.

The citation-only condition shows what the reference carries on its own. Given nothing but the citation, citation-aware retrieval still reaches 62.71 recall@5 on citation-specified questions and 65.54 on content-bearing ones, whereas every learned system stays below 11. The reference is not sufficient—62.71 is well short of 83.62, and Section 5.12 showed why, since the cited number occurs in more than one Act for 92.6% of matched queries—but it carries most of the retrievable signal, and among the systems tested here only explicit structural indexing exploits it effectively.

The two signals are complementary rather than alternative, and the intervention shows how unevenly they are distributed across systems. On content-bearing questions, first-stage retrieval shows no statistically detectable sensitivity to the manipulation: deleting the citation moves BM25 by -2.02, the dense retriever by $+5.40$ and the hybrid by -0.68 recall@5, and corrupting it moves them by -0.67, -3.38 and -4.06, none of these contrasts reaching significance after Holm correction. This is a non-detection rather than a demonstration of independence. Cross-encoder re-ranking does show sensitivity: corrupting the citation costs BM25 + CE -10.14 recall@5 ($p = 0.021$) and Hybrid + CE -10.82 ($p = 0.004$), which is unsurprising once one notes that the re-ranker reads the altered section number as part of the query string. Citation-aware retrieval on the same questions falls much further, from 92.57 to 42.57 when the citation is deleted and to 11.49 when it is corrupted, both at $p < 10^{-18}$. The ordering is the finding: content-based first-stage retrieval is not measurably dependent on the reference, a learned re-ranker depends on it weakly, and explicit structural indexing depends on it almost entirely.

Two consistency checks fall out of the design rather than being imposed. When the citation is deleted, citation-aware retrieval reproduces BM25 exactly (1.13 and 42.57 in the two groups), because with nothing to parse the system is BM25. On the 19 citation-free questions it likewise equals BM25 at every cutoff (63.16), and deleting citations there changes nothing, since there was none to delete.

One limit should be stated plainly. On citation-specified questions every learned system sits between 0.00 and 2.26 recall@5 in all four conditions, so their insensitivity to the intervention there is a floor effect rather than evidence of robustness: they never retrieved these questions to begin with. The finding of no statistically detectable first-stage sensitivity to the reference rests on the content-bearing group, where those systems do work. Even there, the strongest learned configuration reaches 60.14 recall@5 against 92.57 for the structural index, and on citation-specified questions cross-encoder re-ranking reaches 0.56: a learned re-ranker cannot compensate for a reference it has no mechanism to resolve.

5.14 Comparison with Open LLM Baselines

Table 23 compares the best fine-tuned encoder with three open 7–8B LLMs under zero-shot and 3-shot prompting. To ensure a fair comparison, all LLMs are prompted as extractive QA systems. They are instructed to copy the answer verbatim from the supplied statutory paragraph, without paraphrasing or explanation, and to return only the extracted span. Both the encoder and the LLMs therefore perform the same extractive task on the same gold paragraph, so the comparison isolates supervised fine-tuning against prompting rather than differences in output format.

Table 23 Fine-tuned C′ compared with open 7–8B LLMs under zero-shot (ZS) and 3-shot (FS) prompting.

<table>
<tr><th>System</th><th>Params (B)</th><th>ZS F1</th><th>FS F1</th></tr>
<tr><td>Mistral:7b</td><td>7.00</td><td>67.11</td><td>68.83</td></tr>
<tr><td>Llama 3.1:8b</td><td>8.00</td><td>74.00</td><td>76.94</td></tr>
<tr><td>Qwen2.5:7b</td><td>7.00</td><td>77.59</td><td>76.39</td></tr>
<tr><td>C′</td><td>0.11</td><td colspan="2">82.33</td></tr>
</table>

Parameter counts are approximate. The Llama row uses the reproducible `llama3.1:8b` Ollama tag. All LLM baselines are prompted only; none is fine-tuned on IndicIPR-QA.

C′ outperforms the best zero-shot baseline Qwen2.5:7b by 4.74 F1 while using about one sixty-fourth as many parameters. Three-shot prompting gives mixed results: Llama 3.1 improves by 2.94 F1, whereas Qwen2.5 drops by 1.20 F1. This suggests that few-shot prompting alone does not reliably improve verbatim span extraction from statutory text. Across all settings, the compact fine-tuned encoder outperforms the much larger prompted LLMs.

5.15 Error Analysis

A rule-based lexical error analysis was performed on the 178 C′ predictions that do not exactly match the gold answer. Normalised containment identifies boundary errors, predictions within two words of the gold answer are labelled minor differences, and the remaining cases are classified by unique-token overlap. These categories describe lexical differences only and are not human judgments of legal correctness. As shown in Table 24, minor differences and boundary errors account for 62.4% of all non-EM predictions, while only two predictions have no token overlap with the reference.

Table 24 Error categories for EM-wrong predictions of C′ (n=178).

Error type	Count	%
Minor word difference	60	33.7
Span boundary error	51	28.7
Low overlap	34	19.1
Partial overlap	31	17.4
Zero token overlap	2	1.1

Table 25 Lexical error profile across non-EM predictions.

Model	n	Boundary + Minor (%)	Low + Partial (%)	Zero Overlap (%)
A	183	57.9	37.7	4.4
D	185	57.3	35.7	7.0
E	181	64.1	31.5	4.4
G	190	64.7	34.7	0.5
B′	181	64.6	32.0	3.3
D2′	179	57.0	41.3	1.7
E2′	181	64.1	34.3	1.7
F′	176	63.1	33.0	4.0
C′	178	62.4	36.5	1.1
H′	179	63.1	36.3	0.6
I′	182	62.6	36.8	0.5

Table 25 shows that the same pattern holds across all 11 models. Boundary and minor differences account for most non-EM predictions, whereas zero-token-overlap cases remain rare, particularly for the DAPT variants. Most non-exact-match predictions therefore differ from the reference through boundary or small lexical discrepancies rather than complete answer mismatch. These categories describe surface-form differences only. Because this analysis is based only on lexical overlap, it cannot determine whether the predicted spans are legally equivalent to the reference. That would require a separate human evaluation.

5.16 Qualitative Analysis

The previous analyses quantify the main error types. Table 26 illustrates them with three verbatim examples from the test split: one correct extraction, one boundary error, and one retrieval failure. Together, they show why span boundaries matter in legal QA and why retrieval, rather than the reader, limits end-to-end performance.

Case (a) shows a successful extraction. Because the question refers directly to a specific clause, the answer corresponds to that clause and the reader copies it exactly.

Case (b) illustrates the most common error type. The gold answer is clause (b) alone, but the model also includes clauses (a) and (c). Although every returned word comes from the statute, the expanded span changes the legal meaning by extending the set of people covered by the provision. Despite this, token overlap remains 0.61, showing that F1 alone does not fully reflect the importance of span boundaries in legal text.

Case (c) shows a retrieval failure. With the gold paragraph, the reader produces the correct answer. When retrieval is required, however, the system selects a Patents Act paragraph instead of the correct Trade Marks Act paragraph, and the reader extracts a fluent but irrelevant answer. Because the output gives no indication that the wrong statute was retrieved, retrieval errors are silent and can appear convincing. This example illustrates the retrieval–reading tension discussed in Section 5.9 and highlights the importance of citation-aware retrieval.

Table 26 Three verbatim cases from the test split. Outputs are lower-cased and whitespace-tokenised by the span decoder.

(a) Correct extraction. Trade Marks Act, Section 15	
Question	What is laid down in clause (b) of Section 15?
Gold span	statement of number, price, quality or names of places
C′ output	statement of number, price, quality or names of places
Outcome	Exact match.
(b) Boundary error. Geographical Indications Act, Section 76	
Question	Under Section 76, what is stated about agents?
Gold span	a person registered in the prescribed manner as a geographical indications agent, or
C′ output	a legal practitioner, or (b) a person registered in the prescribed manner as a geographical indications agent, or (c) a person in the sole and regular employment of the principal.
Outcome	F1 0.61, exact match 0. The span over-runs into the adjacent clauses, which changes which persons the provision covers.
(c) Retrieval failure. Trade Marks Act, Section 15	
Question	In Section 15, what does clause (d) say?
Gold span	colour, seeks to register those trade marks, they may be registered as a series in one registration.
Gold context	colour, seeks to register those trade marks, they may be registered as a series in one registration. (F1 1.00)
BM25 top-5	that the subject of any claim of the complete specification is not an invention within the meaning of this act ; (F1 0.00)
Outcome	The retrieved paragraph belongs to the Patents Act. The reader answers a Trade Marks question from the wrong statute, with no signal that it has done so.

5.17 Dataset Quality Ablation

The experiments above use the clean curated dataset. To assess whether filtering and manual curation influence downstream performance, a model-level dataset quality ablation was conducted. The architecture, validation set, test set, training schedule, and evaluation metric were kept fixed, with only the training data varying across conditions. All models use InLegalBERT as the base encoder and are evaluated on the same 344-question clean test split. To reduce the effect of random initialisation, each condition is repeated with three random seeds (7, 42, and 99).

The clean curated condition uses the official 3,710-example training split. The remaining conditions are reconstructed from the 5,721 raw GPT-4o-generated candidates using only training-split sections to prevent leakage. Heuristic all keeps all examples passing the nine automated filters but omits manual curation. Three additional conditions disable F3, F4, or both, because these filters removed the largest number of candidates during dataset construction. The Raw candidates condition keeps only the two basic validity checks (F1 and F2), which enforce non-empty fields and verbatim answer spans.

Table 27 Dataset quality ablation. Values are mean ± SD over seeds 7, 42, and 99.

Training variant	Train n	EM (%)	F1 (%)
Clean curated	3,710	47.19 ± 1.02	79.43 ± 0.30
Heuristic all	3,931	44.48 ± 0.77	75.63 ± 0.30
Without F3	4,232	44.48 ± 0.88	77.87 ± 0.64
Without F4	4,268	44.86 ± 1.31	77.51 ± 0.41
Without F3 and F4	4,745	46.61 ± 0.93	79.61 ± 0.52
Raw candidates	4,766	47.09 ± 0.29	79.71 ± 0.49

The clean curated condition repeats the configuration of model A in Table 8 on identical data and the same three seeds, but as an independent set of training runs. The two disagree by 0.34 to 0.91 F1 per seed (79.43 ± 0.30 here against 79.70 ± 0.97 there), which bounds run-to-run reproducibility on this CPU setup at roughly 0.5 F1 and is a further reason to treat sub-point differences between pipelines as noise.

Table 27 shows that Heuristic all is the weakest condition, with a mean F1 of 75.63, indicating that automated filtering alone is insufficient for reliable model training. Removing only F3 or only F4 improves performance but remains below the clean curated baseline, suggesting that relaxing individual filters introduces additional noise without enough useful training examples. In contrast, disabling both F3 and F4 or using the larger raw-candidate pool produces performance similar to the clean curated condition. Because these results are based on only three seeds and no paired statistical comparison, the small differences among these conditions should not be interpreted as evidence that weaker filtering is superior.

Stated directly, the nine automated filters and the manual curation step do not measurably improve training utility on this benchmark: the raw candidate pool trains as well as the curated split, within the resolution of three seeds. Their demonstrated value lies primarily in producing a more carefully validated evaluation set and enabling more interpretable error analysis, rather than in improving training utility. The ablation therefore separates two roles of data curation: training utility and evaluation reliability.

The findings have two implications. For benchmark construction, manual curation remains important because it produces a more reliable evaluation set and enables clearer error analysis. For model training, the interaction between filtering rules is not monotonic, so the effects of weaker filtering should be confirmed with larger multi-seed studies before drawing stronger conclusions.

Overall, the experiments show that adapting InLegalBERT beyond direct fine-tuning produces small numerical gains in extractive QA. Among the four adapted pipelines, C′ achieves the highest F1, although the differences are not statistically significant. The experiments do not identify whether those gains arise from SQuAD transfer, from DAPT, from corpus composition, or from training variability: the largest gains accrue to the encoders least aligned with the task at initialisation, which is consistent with SQuAD transfer supplying general extractive capability, but the design does not separate that explanation from the alternatives (Section 7.4). The remaining challenges are accurate span boundaries and, more importantly, retrieval of the correct statutory paragraph.

6 Generalisation and Robustness Analysis

This section tests whether the adapted pipelines remain reliable beyond the main benchmark results. It examines split separation, unseen legal vocabulary, possible exposure during domain-adaptive pretraining, and the role of synthetic text. The aim is to determine whether the results reflect genuine generalisation rather than data leakage or artifacts of the training data.

6.1 Split-Level Context Separation

The train, validation, and test splits were created at the statute-section level. After paragraph text was normalised, they contained 898, 104, and 113 unique contexts, respectively, with no overlap between any pair of splits. Thus, no paragraph used in supervised training appears in the validation or test sets, and the model is evaluated on questions from previously unseen statute sections. Possible exposure during unsupervised pretraining is examined below.

6.2 Novel Answer Vocabulary

Vocabulary overlap was measured using lowercased alphanumeric tokens, the regular expression `[a-z0-9']+`, and one count for each distinct token in each field. Of the 1,255 distinct answer tokens in the test split, 137 (10.9%) do not appear in any training answer. Similarly, 69 of 467 question tokens (14.8%) and 155 of 1,439 context tokens (10.8%) are unseen in training. Overall, about one test token in nine is unseen during training, so the models must generalise to legal terminology that they did not encounter during training.

6.3 What the Split Controls Establish

Context overlap between splits is zero for all Acts, and the novel-vocabulary analysis shows that evaluation is not a repeat-context test. These controls reduce the risk of direct memorisation, but they do not prove that the observed differences are caused by domain adaptation. BM25 is not a memorisation baseline: it must retrieve a paragraph, whereas the reader is given the gold paragraph. The F1 gap therefore compares different task setups and is not evidence about learned representation quality.

One caveat concerns the DAPT corpus. The masked language model stage used an archived authentic statutory pool comprising 898 training-split contexts and three audited test contexts, together with synthetic passages generated from training-split seed entities. An audit of the archived corpora found no validation context in any

DAPT file and no evaluation context of either kind in the synthetic-only corpus. Three test contexts are present in the authentic-only and mixed corpora, supplying the context for six of the 344 test questions (1.74%). This exposure is unsupervised because the masked language model stage uses neither questions nor answer spans. The three contexts are also absent from the training split, so they are genuine evaluation material rather than statutory text repeated across sections.

The archived corpus records how the exposure arose. The authentic block contains exactly 3,710 records, one per training QA pair, which is the size of the training split; these collapse to 901 distinct passages after normalisation. The corpus was therefore built against the section-level split as it stood when the corpus was assembled. Measured against the final split, 898 of those passages are training contexts and three are test contexts: Patents section 128, and Trade Marks sections 106 and 110. Each of the three contributes exactly two corpus records and exactly two test questions. That correspondence is what would be expected if three whole sections were reassigned from the training to the test side when the split was regenerated, because section-level splitting keeps every item from a section on the same side; it is not consistent with the piecewise inclusion of individual evaluation items. The archived corpus was not rebuilt against the final split, so the three contexts remain in it. Every figure in this paragraph can be recomputed from the archived corpus and the released splits.

The effect can be measured directly. Excluding the six affected questions, C' scores 48.52 EM and 82.22 F1, compared with 48.26 EM and 82.33 F1 on the full test set. This is a change of +0.26 EM and −0.11 F1. Exact match increases when the affected questions are removed, so the reported aggregate is not inflated by them. The held-out perplexity in Table 10 has the same caveat and should be understood as measured on mostly, but not entirely, unseen statutory text.

6.4 Are the Gains an Artifact of Synthetic Text?

The benchmark was generated with GPT-4o and the DAPT corpus with Gemma 3. Thus, two layers of machine-generated text underlie these results. This raises a fair question: does the encoder become better at reading Indian statutes, or only at reading LLM-written legal prose? Three properties of the design address this question.

First, every evaluation context is authentic statutory text, and every answer is an exact span of that text. The questions were machine-generated, then filtered and manually curated, so question-style artifacts may remain. Gemma-generated passages are used only in the unsupervised MLM stage.

Second, the split is section-level, and the vocabulary check above shows that roughly one test answer token in nine does not appear in any training answer. The gains therefore have to survive both unseen statutory sections and unseen legal terminology.

Third, the synthetic-only variant I' is the strongest available test of this question. Its encoder sees no authentic statutory text during DAPT, only 13,211 generated passages, yet reaches 81.86 F1 on authentic test contexts. This is numerically higher than direct fine-tuning (80.56) and the authentic-only DAPT variant H' (81.43). However, the differences are not statistically separable. The result therefore supports the feasibility of transfer, not its superiority.

The concern nevertheless remains. The synthetic corpus was validated with rule-based checks rather than by legal experts, so generation artifacts may remain. In addition, the effect sizes are small enough that the present test set cannot distinguish the DAPT variants. The evidence supports a narrower claim: the synthetic-only condition shows that generated statutory-style text can transfer to authentic statutory contexts without evaluation-context exposure during DAPT. Its magnitude was addressed by the size-matched, multi-seed experiment in Section 5.5, which finds no composition effect above seed noise; the mechanism remains open.

7 Discussion

This section interprets the results in four steps. It explains the differences between the encoder families, examines why the Plant Varieties Act is hardest, discusses the implications for legal practice and system design, and then states the study's limitations and future work.

Three boundaries apply to everything that follows, and are stated here rather than left to Section 7.4 so that the claims are read at their intended scope. First, the structural diagnosis is a finding about IndicIPR-QA and about statutory QA settings with a comparable density of citation-specified questions; it is not a general claim about legal retrieval, and a benchmark whose questions describe provisions by content would not exhibit the same bottleneck. Second, the paper does not establish that dense retrieval is inferior to sparse retrieval in general. The dense models tested are general-purpose encoders applied zero-shot, without in-domain training, so the comparison bounds what is available off the shelf rather than what dense retrieval can achieve. Third, the paper does not show that synthetic DAPT is without value. It reports no detectable effect of corpus composition under one controlled setup, at one corpus scale, on one benchmark, with three seeds and a 344-question test set, and Section 5.5 gives the interval within which a real effect could still hide.

7.1 Why Do the Encoder Families Behave Differently?

The direct and transfer baselines separate domain knowledge from general extractive QA skill. Under direct fine-tuning, InLegalBERT leads because it is the closest match to the target setting: an Indian legal encoder applied to Indian statutory text. LEGAL-BERT remains competitive, only 0.81 F1 behind, even though it contains no Indian legal text. Because LEGAL-BERT was pretrained on legal text from several Western jurisdictions (EU, UK, and US), this result suggests that statutory register—clause structure, conditional phrasing, and definitional syntax—transfers across jurisdictions. BERT-base is clearly weakest, showing that general English pretraining alone does not provide enough legal vocabulary or clause-level bias for this task.

InCaseLawBERT presents a useful contrast. It has Indian legal pretraining, yet it falls behind both InLegalBERT and LEGAL-BERT. The likely explanation is a genre effect. Court judgments are discursive, narrative, and argument-driven, whereas the benchmark contexts are definitional and provision-like. They require the model to reproduce long, tightly bounded statutory clauses rather than to locate reasoning spans in case-law prose.

SQuAD v1 transfer mainly acts as a general QA warm-up. The largest gains go to BERT-base and InCaseLawBERT, the two encoders least aligned with extractive statute QA at initialisation. The intermediate task therefore seems to teach span-extraction behaviour: where answers start and end, how to handle overlapping candidates, and how to set boundaries for longer answers. InLegalBERT gains only 0.89 F1 from the same stage, suggesting that its main advantage comes from stronger legal representations before intermediate training begins.

The InLegalBERT routes form a narrow band. B' is 0.89 F1 above A, while H', I', and C' are 0.87, 1.30, and 1.77 points above A. These differences do not reveal the mechanism of improvement, for the reasons discussed in Section 7.4. Authentic text may support statutory register, synthetic text may broaden vocabulary, and SQuAD transfer may improve span selection. Section 5.5 finds no detectable composition effect at the resolution of the present experiment, since authentic, synthetic, and mixed corpora of equal scale are statistically indistinguishable across the three tested seeds. What text composition does to the representation, as opposed to how much it moves span F1, would need a factorial experiment with probing.

7.2 Why Is the Plant Varieties Act So Hard?

The Plant Varieties and Farmers' Rights Act is the hardest Act (F1 = 63.27%), even though its answers are the shortest, at 21.9 words on average. One possible explanation is its biological terminology, including "essentially derived variety," "propagating material," and "breeder," together with high question–context lexical overlap (0.636). These features may create several plausible spans with similar surface cues. The present experiments do not test whether terminology or another Act-level property causes the gap. Act-targeted DAPT is therefore a testable follow-up, not a demonstrated remedy (Section 7.4).

7.3 Implications for Legal Practice and System Design

The results suggest three practical points for anyone building or buying a statutory question-answering system for Indian IPR law.

First, verifiability is affordable. A 110M-parameter encoder restricted to copying contiguous spans reached 82.33 F1 on gold paragraphs, above every 7–8B model prompted to extract answers verbatim. Because its output is a substring of a named provision, a lawyer can check it against the statute quickly, and the system cannot invent a plausible rule that does not appear in an Act. This is valuable for due diligence, prior-art screening, and first-pass statutory research, where verifiability matters more than fluency. The model also runs on a desktop CPU, which may matter to practitioners who handle privileged material and cannot send queries to an external service.

Second, the main practical risk is a silent retrieval failure. The same reader that answers correctly from the right paragraph falls to 30.00 F1 when it must find that paragraph itself. Table 26(c) shows what a user may see: a fluent answer drawn from the wrong Act, with no warning that it is unreliable. A deployed system should therefore display the retrieved provision with every answer and provide a retrieval-confidence signal. Users should check the cited provision, not only the generated prose. Answer-only interfaces are unsafe in this setting, regardless of reader quality.

Third, boundary precision is a legal issue, not a cosmetic one. In Table 26(b), the model crosses a clause boundary and changes who may act as a registered agent, while still scoring 0.61 token F1. Aggregate overlap metrics understate this problem because they treat a proviso or adjacent sub-clause as a few extra tokens rather than as a condition that changes the rule. Evaluations used for procurement should report exact match and inspect boundary errors directly. Automatic extraction should also receive clause-level review before anyone relies on the output.

More broadly, the difficulty ordering across Acts appears to be a property of the statutes rather than the model, and it is stable across all eleven pipelines. Acts whose vocabulary differs most from general legal language, such as Plant Varieties, will need targeted resources. Buyers should not assume that performance on Patents or Designs will transfer to every other Act.

7.4 Limitations and Future Work

This study has three main limitations: the strength of the statistical evidence, the design of the DAPT comparison, and the scope of the benchmark and baselines. Each limitation suggests a next step.

The first limitation is statistical power. The test set contains 344 questions. Paired bootstrap testing with Holm correction shows that the +1.77 F1 advantage of C' over direct fine-tuning (A) is not significant (p=0.218). The differences among C', B', H', and I' are also not significant (p=0.690). The three-seed repetition in Table 8 covers only A and C' and reuses the DAPT and SQuAD-transfer checkpoints. Its spread is therefore a lower bound on total training variation. The effect is positive in all three seeds, which supports its direction but not its exact size: the paired mean of +1.37 F1 has SD = 0.55 over n=3. Seed 42 is the best seed for both systems, so the single-seed values reported elsewhere are optimistic. Seeds covering the full pipeline are available for the size-matched corpora (Table 12), where varying the seed across all three stages gives a pooled within-corpus SD of 0.53 F1; the corresponding figure for A and C' at their published scales was not affordable within the CPU-only budget, nor was a larger test set.

The second limitation concerns the DAPT comparison. The three corpora are not the same size. Table 5 gives 3,710 records over 901 distinct passages for H', 14,214 for C', and 13,211 for I'. The archived authentic pool contains 901 distinct paragraphs, comprising 898 distinct normalised training contexts and three audited test contexts, so matching the much larger synthetic pool would require substantial upsampling of this authentic pool and could introduce an additional artifact. This problem affects comparisons among H', C', and I', but not the comparison between C' and A, because A has no DAPT corpus.

Checkpoint selection also differs. C' and I' keep the checkpoint with the lowest validation loss, while A, the SQuAD-transfer baselines, and H' keep the checkpoint with the highest validation F1. Inspecting the retained checkpoints shows that this matters less than the description suggests. Every run trains for exactly three epochs with evaluation after each epoch, so the rule chooses among three candidates. For A the two rules select the same epoch, and evaluating both selections directly gives identical scores of 46.80 EM and 80.56 F1. The comparison between A and C', on which the multi-seed analysis rests, is therefore unaffected. For C' and I' the loss rule retained the final epoch, so no later checkpoint was discarded. H' is the only model where the rules demonstrably diverge, because its validation loss is lowest after the first epoch while the F1 rule retained the third. Re-training H's final stage under the loss rule confirms both the divergence and its size. The loss rule again selects the first epoch and gives 47.97 EM and 81.00 F1, against 47.97 and 81.43 for the published F1-selected checkpoint. Unifying the rule across A, H', C', and I' therefore moves one model by 0.43 F1, leaves exact match unchanged, and preserves the ordering C' above I' above H' above A. The SQuAD-transfer baselines retain F1 selection and lie outside this comparison.

Section 5.4 addresses both confounds directly, holding corpus scale, record multiplicity, and checkpoint selection fixed while varying corpus composition, subject to the small asymmetric evaluation-text exposure documented below. No single factor accounts for the adaptation gain at a magnitude this setup can resolve. Composition at fixed scale spans 0.73 F1, a fifteen-fold increase in scale is worth 0.67 F1, record multiplicity is worth 0.92 F1, and the checkpoint rule moves one model by 0.43 F1, against a seed-to-seed standard deviation of about 1.0 F1. Synthetic text does not outperform authentic text at equal scale. The design therefore shows a small numerical gain from the adapted InLegalBERT pipeline across the tested final fine-tuning seeds, while the source and magnitude of that gain remain uncertain, and it also shows that attributing that gain to synthetic text specifically is not warranted. Section 5.5 closes the seed confound for the first of these quantities by repeating the whole pipeline three times per matched corpus: the composition effect falls to 0.08 F1 between means against a pooled within-corpus SD of 0.53, analysis of variance detects no corpus-composition effect ($p = 0.98$), and the seed-42 ordering reverses. The power of that test is nevertheless limited. With three seeds the pairwise 95% confidence intervals reach ± 2.7 F1, wider than the $+1.77$ F1 attributed to adaptation overall. The experiment therefore provides no evidence that corpus composition explains the observed adaptation gain at the resolution of the present test set and seed budget: the observed composition differences are small relative to training variability and do not reproduce consistently across the three seeds tested. Scale, record multiplicity, and the checkpoint rule have not been repeated and remain single-seed bounds. These limitations do not affect the retrieval results in Section 5.9.

The authentic-only and mixed corpora also contain evaluation text. H' and C' each contain three test contexts covering six test questions; among the size-matched corpora, auth901 contains the same three test contexts covering six questions, mixed901 contains one covering two, and synth901 contains none. The measured effect on C' is -0.11 F1, and removing those questions raises exact match. These pipelines are therefore not completely free of evaluation text during pretraining. The exposure is unsupervised, confined to the MLM stage, and asymmetric across the size-matched corpora, but it does not account for the composition result in Section 5.5: auth901 carries the most test text and records the lowest mean F1 of the three, so the ordering runs opposite to what leakage-driven inflation would produce.

The third limitation concerns the baselines. The 7–8B LLMs are evaluated only with zero-shot and 3-shot prompts. The evidence therefore shows that a compact adapted encoder beats larger prompted models on verbatim span extraction; it does not show that it would beat them after equal supervised training. Accordingly, Table 23 should be interpreted as a comparison of supervised compact extraction against prompting-only large-model extraction, not as a parameter-scale comparison. Fine-tuning these models on IndicIPR-QA would test that directly.

The dense retrievers in Section 5.9 are also general-purpose checkpoints used zero-shot. None was trained on legal text or on statutory question–paragraph pairs. Their failure to displace BM25 does not rule out a legal-domain retriever trained for this task, which remains the most important untested retrieval baseline and should be evaluated in future work. Encoding is deterministic, but the comparison still uses only one BM25 implementation and four public checkpoints. All four are also small, at roughly 33 million parameters, so the finding is that these accessible CPU-friendly checkpoints do not displace BM25, not that dense retrieval in general cannot. Larger encoders, late-interaction models, and query expansion remain untested. That gap matters less than it would otherwise, because Section 5.12 shows that a major source of the observed ceiling is structural rather than something that requires greater semantic model capacity to overcome: a retriever with no learned component at all reaches 86.34% recall@5 where the best learned system reached 30.23%.

The citation-aware result carries its own limits. It works because 94.48% of IndicIPR-QA questions contain a parsable citation, which is a property of this benchmark rather than of statutory QA in general. Only 19 test questions lack one, so the 63.16% recall@5 measured on that subset is imprecise and cannot support a claim about how the method would behave on citation-free queries. A deployment in which users describe a problem in their own words would depend on the content-bearing behaviour of the underlying retriever, which is exactly the regime this benchmark under-samples. Testing that properly needs a benchmark whose questions identify provisions by content.

Cross-encoder re-ranking is no longer untested here, and Section 5.11 shows it gives the largest improvement among learned systems, but it is bounded by a first-stage recall@50 of about 42%. The remaining directions all act on candidate generation rather than on re-ordering: query rewriting that expands a citation into its governing text, metadata-aware indexing over sub-section and clause structure as well as section number, and first-stage retrievers trained on legal question–paragraph pairs.

Finally, the benchmark has a limited scope. Its 4,462 pairs are all answerable from a single section, so it mainly tests span extraction rather than legal reasoning. Models are never penalised for always predicting a span. Some questions cite a section without explaining what it concerns, which makes reading easy and retrieval nearly impossible. The Designs Act shows this problem with 99.42 reader F1 and 0.00% recall@5, so gold-context scores for these Acts are an upper bound. The questions were machine-generated and then curated, while the synthetic corpus was checked only with rules. Some generation artifacts may therefore remain. Coverage is limited to the main statutory text in English.

The most useful extensions are unanswerable, scenario-based, multi-section, and paraphrased questions; questions that identify their provision by content; and Indian-language versions. Act-targeted DAPT with additional Plant Varieties passages remains a testable remedy for the hardest Act.

8 Conclusion

This paper set out to explain why extractive QA over Indian IPR statutes appears much harder end to end than at either stage alone. Gold-context span extraction is not the dominant source of end-to-end error: given the governing paragraph, the supervised encoder reaches 82.33 F1; under the separate prompting-only comparison in Section 5.14, it also scores above the tested 7–8B models. This is not a claim that the reader is solved. Boundary errors and near-misses account for 62.4% of its non-exact-match predictions (Table 24), so token F1 conceals span disagreements that a reader of a statute would notice. The point is one of relative magnitude. Retrieval appeared to be the limitation, with end-to-end F1 of 30.00, BM25 recall@5 of 22.67%, no consistent gain from four zero-shot dense retrievers, and only 30.23% from a cross-encoder re-ranker bounded by a first-stage ceiling near 42%.

The analysis shows that much of this retrieval difficulty on IndicIPR-QA is structural: citation-specified questions require explicit reference resolution that ordinary content-based retrievers do not provide. Half the test set identifies its target by citing a section rather than by describing content. Those items are easy to extract from once supplied and nearly impossible to retrieve, which accounts for the inversion in which the Act with the highest gold-context score has the lowest retrieval recall. Indexing statutory structure removes most of the gap: a regular expression mapping cited sections onto passage metadata, with nothing learned, raises recall@5 from 22.67% to 86.34% and recall@10 to 95.64%, well past what any re-ranker over the same candidate pool could reach. Content-bearing questions were already at 46.71% recall@5 under plain BM25, against 0.00% for citation-specified questions. The reported gold-context score decomposes accordingly, into 88.63 F1 on the citation-specified half and 75.65 on the content-bearing half.

Intervening on the reference confirms the mechanism. On the 177 citation-specified questions, deleting the citation reduces citation-aware recall@5 from 83.62% to 1.13%, and replacing it with another valid section of the same Act reduces it to 0.00%, while first-stage content-based retrieval shows no statistically detectable change. Citation-only queries still retain 62.71%, so structural reference information is dominant but not sufficient without content-based disambiguation. Statutory retrieval on this benchmark is driven jointly by substantive content and explicit legal-reference structure; ordinary first-stage retrieval primarily captures the former, while explicit indexing makes the latter directly and reliably available to the pipeline.

Synthetic corpus composition contributes no detectable advantage once it is measured under size-matched, multi-seed control. With corpus scale, record multiplicity, and checkpoint selection held fixed, synthetic text does not outperform authentic text, and every effect is smaller than seed-to-seed variation. Repeating the full pipeline under three seeds for each matched corpus separates effect from noise directly: the corpus means span 0.08 F1 against a pooled within-corpus standard deviation of 0.53, analysis of variance finds no detectable corpus effect, and the ordering seen at a single seed reverses. This is reported as a controlled non-detection; it does not establish equivalence between corpus compositions.

Two conclusions follow. For citation-heavy statutory QA settings such as IndicIPR-QA, structural reference resolution—through metadata-aware indexing, citation parsing, and query rewriting—should be addressed before further optimisation of the semantic retrieval and synthetic-pretraining approaches tested here. And benchmarks should ensure that every question identifies its provision by content: a citation may be retained, but it should accompany substantive content rather than replace it, because items that supply only a citation inflate gold-context scores, depress retrieval scores, and measure neither capability well.

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Declarations

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- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** All source texts are official Indian government legislative documents freely available from government portals, and contain no personally identifiable information. The human validation study involved adult annotators rating non-personal statutory question–answer items; participation was voluntary, informed consent was obtained, and annotators were free to withdraw at any point. No intervention was performed and no personal data were collected or retained. *[Institutional ethics determination and reference number to be inserted; withheld here as identifying.]*
- **Data availability.** An anonymous archive supporting this submission is available at https://anonymous.4open.science/r/Synthetic-DAPT-Paper-5646. It contains the question–answer splits used here (3,710 training, 408 validation, and 344 test pairs, in both the native and SQuAD-style formats) and the three archived domain-adaptive pretraining corpora, so that the corpus manifest in Table 5 and the split- and pretraining-level context audits reported in Section 6 can be recomputed independently. A SHA-256 manifest covers every file. The public archival citation and DOI will be supplied after anonymous review.
- **Code availability.** The same archive contains the full pipeline: seed extraction and synthetic passage generation, corpus assembly, masked language model pretraining for all three DAPT variants, SQuAD v1 transfer, final extractive QA fine-tuning

for all evaluated encoder configurations, span prediction and scoring, the prompted LLM baselines, the retrieval experiments, and figure generation, together with a pinned environment specification. The public repository URL will be supplied after anonymous review.

The dataset and the models reported here are intended for research purposes only. Neither should be used as a substitute for professional legal advice or as a definitive statement of Indian intellectual property law.

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